

Physics Publications with Major Personal Contributions

Oliver Gutsche

May 10, 2026

1. Search for a new heavy scalar resonance decaying into the Higgs boson and a new scalar particle in the $b\bar{b}b\bar{b}$ final state using proton-proton collisions at $\sqrt{s} = 13$ TeV

Hayrapetyan, Aram et al.

[arXiv:2605.02848](#) | 2026-05-04

2. Observation of the rare decay $\eta \rightarrow \mu^+ \mu^- e^+ e^-$

Hayrapetyan, Aram et al.

[arXiv:2605.00615](#) | 2026-05-01

3. Search for light charged Higgs bosons decaying to charm and strange quarks in $t\bar{t}$ events in proton-proton collisions at $\sqrt{s} = 13$ TeV

Hayrapetyan, Aram et al.

[arXiv:2604.27795](#) | 2026-05-01

4. Improved results on Higgs boson pair production in the $4b$ final state

Hayrapetyan, Aram et al.

[arXiv:2604.27044](#) | 2026-04-29

5. Search for electroweakinos in compressed-spectrum scenarios with low-momentum isolated tracks in proton-proton collisions at $\sqrt{s} = 13$ TeV

Hayrapetyan, Aram et al.

[arXiv:2604.25604](#) | 2026-04-28

6. Simultaneous measurements of N-subjettiness observables in jets from gluons and light-flavour quarks, and in decays of boosted W bosons and top quarks

Hayrapetyan, Aram et al.

[arXiv:2604.25538](#) | 2026-04-28

7. Measurement of the $Z \rightarrow \mu^+ \mu^-$ angular coefficients in pp collisions at $\sqrt{s} = 13$ TeV as functions of transverse momentum and rapidity

Gevorgyan, Arzunik et al.

[arXiv:2604.25678](#) | 2026-04-28

8. Search for associated production of a Higgs boson and two vector bosons via vector boson scattering at $\sqrt{s} = 13$ TeV

Hayrapetyan, Aram et al.

[arXiv:2604.24531](#) | 2026-04-27

9. Search for dark matter produced in association with a dark Higgs boson decaying into a bottom quark-antiquark pair in proton-proton collisions at $\sqrt{s} = 13$ TeV

Hayrapetyan, Aram et al.

[arXiv:2604.20999](#) | 2026-04-22

10. Search for the single production of vector-like quarks decaying into a W boson and a b quark using single-lepton final states in proton-proton collisions at $\sqrt{s} = 13$ TeV

Hayrapetyan, Aram et al.

[arXiv:2604.17564](#) | 2026-04-19

11. Highly boosted dielectron identification in proton-proton collisions at $\sqrt{s} = 13$ TeV

Hayrapetyan, Aram et al.

[arXiv:2604.13320](#) | 2026-04-14

12. Search for heavy resonances decaying into four-lepton final states via light bosons in proton-proton collisions at $\sqrt{s} = 13$ TeV

Hayrapetyan, Aram et al.

[arXiv:2604.14236](#) | 2026-04-14

13. A search for microscopic black holes, string balls, and sphalerons in proton-proton collisions at $\sqrt{s} = 13$ TeV

Hayrapetyan, Aram et al.

[arXiv:2604.10732](#) | 2026-04-12

14. Search for a new heavy resonance decaying to a top quark and a neutral scalar boson in proton-proton collisions at $\sqrt{s} = 13$ TeV

Hayrapetyan, Aram et al.

[arXiv:2604.10729](#) | 2026-04-12

15. Particle transformers for identifying Lorentz-boosted Higgs bosons decaying to a pair of W bosons

Hayrapetyan, Aram et al.

[arXiv:2604.09809](#) | 2026-04-10

16. Observation of nuclear suppression in coherent $\Upsilon(1S)$ photoproduction off heavy nuclei at the LHC

Belyaev, Andrey et al.

[arXiv:2604.05814](#) | 2026-04-07

17. Search for soft unclustered energy patterns produced in association with a W or Z boson in proton-proton collisions at $\sqrt{s} = 13$ TeV

Hayrapetyan, Aram et al.

[arXiv:2604.05996](#) | 2026-04-07

18. Evidence of $ZZ\gamma$ production and observation of $4\ell\gamma$ in proton-proton collisions at $\sqrt{s} = 13$ TeV

Hayrapetyan, Aram et al.

[arXiv:2604.02594](#) | 2026-04-03

19. Search for Higgs boson pair production in the $b\bar{b}WW$ decay channel with two leptons in the final state using proton-proton collision data at $\sqrt{s} = 13.6$ TeV

Gevorgyan, Arzunik et al.

[arXiv:2604.02127](#) | 2026-04-02

20. Measurement of dijet angular distributions and search for beyond the standard model physics in proton-proton collisions at $\sqrt{s} = 13$ TeV

Gevorgyan, Arzunik et al.

[arXiv:2603.25458](#) | 2026-03-26

21. Search for new particles decaying into top quark-antiquark pairs in proton-proton collisions at $\sqrt{s} = 13$ TeV

Gevorgyan, Arzunik et al.

[arXiv:2603.23454](#) | 2026-03-24

22. Search for low-mass vector and scalar resonances decaying into a quark-antiquark pair in proton-proton collisions at $\sqrt{s} = 13$ TeV

Hayrapetyan, Aram et al.

[arXiv:2603.21965](#) | 2026-03-23

23. Search for Higgs boson production at high transverse momentum in the WW decay channel in proton-proton collisions at $\sqrt{s} = 13$ TeV

Hayrapetyan, Aram et al.

[arXiv:2603.22233](#) | 2026-03-23

24. Measurement of the jet mass in hadronic decays of boosted W bosons at 13 TeV and extraction of the W boson mass

Hayrapetyan, Aram et al.

[arXiv:2603.19963](#) | 2026-03-20

25. Jet peak shapes based on two-particle angular correlations in lead-lead collisions at $\sqrt{s_{NN}} = 5.02$ TeV

Belyaev, Andrey et al.

[arXiv:2603.14385](#) | 2026-03-15

26. Measurement of the t-channel single top quark cross section in proton-proton collisions at $\sqrt{s} = 5.02$ TeV

Hayrapetyan, Aram et al.

[arXiv:2603.13592](#) | 2026-03-13

27. Search for Z' bosons decaying into charginos in final states with two oppositely charged leptons and missing transverse momentum in pp collisions at $\sqrt{s} = 13$ TeV

Hayrapetyan, Aram et al.

[arXiv:2603.11035](#) | 2026-03-11

28. Measurement of angular correlations inside jets induced by gluon polarization in proton-proton collisions at $\sqrt{s} = 13.6$ TeV

Hayrapetyan, Aram et al.

[arXiv:2603.03689](#) | 2026-03-04

29. Search for a narrow resonance with a mass between 10 and 70 GeV decaying to a pair of photons in proton-proton collisions at $\sqrt{s} = 13$ TeV

Hayrapetyan, Aram et al.

[arXiv:2603.03250](#) | 2026-03-03

30. Combination of ATLAS and CMS searches for Higgs boson pair production at $\sqrt{s} = 13$ TeV

Aad, Georges et al.

[arXiv:2602.23991](#) | 2026-02-27

31. System-size dependence of charged-particle suppression in ultrarelativistic nucleus-nucleus collisions

Belyaev, Andrey et al.

[arXiv:2602.21325](#) | 2026-02-24

32. Search for pair production of heavy resonances in final states with a photon and large-radius jets in proton-proton collisions at $\sqrt{s} = 13$ TeV

Hayrapetyan, Aram et al.

[arXiv:2602.20477](#) | 2026-02-24

33. Observation of the jet diffusion wake using dijets in heavy ion collisions

Hayrapetyan, Aram et al.

[arXiv:2602.19431](#) | 2026-02-23

34. Search for a new resonance decaying to a Higgs boson and a scalar boson in events with two b jets and two Z bosons in proton-proton collisions at $\sqrt{s} = 13.6$ TeV

Hayrapetyan, Aram et al.

[arXiv:2602.18223](#) | 2026-02-20

35. Combined measurements and interpretations of Higgs boson production and decay in proton-proton collisions at $\sqrt{s} = 13$ TeV

Hayrapetyan, Aram et al.

[arXiv:2602.18611](#) | 2026-02-20

36. First measurement of jet axis decorrelation with photon-tagged jets in pp and PbPb collisions at 5.02 TeV

Hayrapetyan, Aram et al.

[arXiv:2602.18279](#) | 2026-02-20

37. Measurement of event shape variables using charged particles inside jets in proton-proton collisions at $\sqrt{s} = 13$ TeV

Hayrapetyan, Aram et al.

[arXiv:2602.17509](#) | 2026-02-19

38. Influence of strangeness on the anisotropic flow of prompt $D^{\pm}_s$ mesons in PbPb collisions at $\sqrt{s_{NN}} = 5.02$ TeV

Belyaev, Andrey et al.

[arXiv:2602.14221](#) | 2026-02-15

39. High-level hadronic tau lepton triggers of the CMS experiment in proton-proton collisions at $\sqrt{s} = 13.6$ TeV

Hayrapetyan, Aram et al.

[JINST 21 \(2026\) P04002](#) | [arXiv:2602.11359](#) | [DOI:10.1088/1748-0221/21/04/P04002](#) | 2026-02-11

40. Measurement of B meson production fraction ratios in proton-proton collisions at $\sqrt{s} = 13$ TeV using open-charm and charmonium decays

Hayrapetyan, Aram et al.

[arXiv:2602.10270](#) | 2026-02-10

41. Probing early parton emissions in heavy ion collisions using the Lund jet plane

Belyaev, Andrey et al.

[arXiv:2602.09271](#) | 2026-02-09

42. Observation of $\Upsilon(1S) + Z$ associated production and measurement of the effective double-parton scattering cross section in proton-proton collisions at $\sqrt{s} = 13$ TeV

Hayrapetyan, Aram et al.

[arXiv:2602.02770](#) | 2026-02-02

43. Observation of a family of all-charm tetraquarks

Hayrapetyan, Aram et al.

[arXiv:2602.02252](#) | 2026-02-02

44. Search for heavy scalar resonances decaying to Lorentz-boosted Higgs and Higgs-like bosons in the $b\bar{b}4q$ final state at $\sqrt{s} = 13$ TeV

Hayrapetyan, Aram et al.

[arXiv:2602.00273](#) | 2026-01-30

45. Measurement and effective field theory interpretation of the photon-fusion production cross section of a pair of W bosons in proton-proton collisions at $\sqrt{s} = 13$ TeV

Hayrapetyan, Aram et al.

[arXiv:2601.21574](#) | 2026-01-29

46. Measurement of the $\Upsilon(1S)$, $\Upsilon(2S)$, and $\Upsilon(3S)$ differential cross sections in pp collisions at $\sqrt{s} = 13.6$ TeV

Hayrapetyan, Aram et al.

[arXiv:2601.20023](#) | 2026-01-27

47. Search for heavy resonances decaying into two Higgs bosons in the $b\bar{b}\tau^+\tau^-$ final state in proton-proton collisions at $\sqrt{s} = 13$ TeV

Hayrapetyan, Aram et al.

[arXiv:2601.20011](#) | 2026-01-27

48. Search for heavy long-lived charged particles with level-1 trigger scouting data from proton-proton collisions at $\sqrt{s} = 13.6$ TeV

Hayrapetyan, Aram et al.

[arXiv:2601.20063](#) | 2026-01-27

49. Strategy and performance of the CMS long-lived particle trigger program in proton-proton collisions at $\sqrt{s} = 13.6$ TeV

Hayrapetyan, Aram et al.

[arXiv:2601.17544](#) | 2026-01-24

50. Search for the pair production of long-lived supersymmetric partners of the tau lepton in proton-proton collisions at $\sqrt{s} = 13$ TeV

Hayrapetyan, Aram et al.

[arXiv:2601.17576](#) | 2026-01-24

51. Full event interpretation with machine-learning-based particle-flow reconstruction in the CMS detector

Hayrapetyan, Aram et al.

[arXiv:2601.17554](#) | 2026-01-24

52. Measurement of the $Z\gamma$ production cross section and search for anomalous neutral triple gauge couplings in pp collisions at $\sqrt{s} = 13$ TeV

Hayrapetyan, Aram et al.

[arXiv:2601.14102](#) | 2026-01-20

53. Combination of searches for heavy vector boson resonances in proton-proton collisions at $\sqrt{s} = 13$ TeV

Hayrapetyan, Aram et al.

[arXiv:2601.12583](#) | 2026-01-18

54. Search for dark matter produced in association with a Higgs boson decaying to bottom quarks in proton-proton collisions at $\sqrt{s} = 13$ TeV

Hayrapetyan, Aram et al.

[arXiv:2601.11330](#) | 2026-01-16

55. Search for a boosted Higgs boson decaying to bottom quark pairs in association with a W or Z boson in proton-proton collisions at $\sqrt{s} = 13$ TeV

Hayrapetyan, Aram et al.

[arXiv:2601.05362](#) | 2026-01-08

56. Measurement of the Higgs boson total decay width using the $H \rightarrow WW \rightarrow e\nu\mu\nu$ decay channel in proton-proton collisions at $\sqrt{s} = 13$ TeV

Hayrapetyan, Aram et al.

[arXiv:2601.05168](#) | 2026-01-08

57. Search for exotic Higgs boson decays $H \rightarrow \mathcal{A}\mathcal{A}$ with $\mathcal{A}\mathcal{A} \rightarrow \gamma\gamma$ in events with a semi-merged topology in proton-proton collisions at $\sqrt{s} = 13$ TeV

Hayrapetyan, Aram et al.

[arXiv:2601.00183](#) | 2026-01-01

58. Search for charged Higgs bosons decaying into top and bottom quarks in lepton+jets final states in proton-proton collisions at $\sqrt{s} = 13$ TeV

Hayrapetyan, Aram et al.

[arXiv:2512.24471](#) | 2025-12-30

59. Machine-learning techniques for model-independent searches in dijet final states

Hayrapetyan, Aram et al.

[arXiv:2512.20395](#) | [DOI:10.5281/zenodo.16656501](#) | 2025-12-23

60. Characterization of the quantum state of top quark pairs produced in proton-proton collisions at $\sqrt{s}=13$ TeV using the beam and helicity bases

Hayrapetyan, Aram et al.

[Phys.Rev.D 113 \(2026\) L051101](#) | [arXiv:2512.17557](#) | [DOI:10.1103/hkgw-dztk](#) | 2025-12-19

61. Measurement of $Z\gamma$ production in proton-proton collisions at $\sqrt{s} = 13.6$ TeV and constraints on neutral triple gauge couplings

Hayrapetyan, Aram et al.

[arXiv:2512.08582](#) | 2025-12-09

62. Measurements of electroweak production of a photon in association with two jets in proton-proton collisions at $\sqrt{s} = 13$ TeV

Hayrapetyan, Aram et al.

[arXiv:2512.00502](#) | 2025-11-29

63. Search for light pseudoscalar bosons, pair-produced in Higgs boson decays in the four-electron final state in proton-proton collisions at $\sqrt{s} = 13$ TeV

Hayrapetyan, Aram et al.

[arXiv:2511.19563](#) | 2025-11-24

64. Search for new physics in the final state with a single photon and large missing transverse momentum in proton-proton collisions at $\sqrt{s}=13$ TeV

Hayrapetyan, Aram et al.

[Phys.Rev.D 113 \(2026\) 072020](#) | [arXiv:2511.17310](#) | [DOI:10.1103/f9r8-styj](#) | 2025-11-21

65. Search for Higgsinos in final states with low-momentum lepton-track pairs at 13 TeV

Hayrapetyan, Aram et al.

[arXiv:2511.16394](#) | 2025-11-20

66. Search for heavy $H\gamma$ and $Z\gamma$ resonances with a bottom quark-antiquark pair in the final state in proton-proton collisions at $\sqrt{s} = 13$ TeV

Hayrapetyan, Aram et al.

[arXiv:2511.14583](#) | 2025-11-18

67. Search for low-mass hidden-valley dark showers with non-prompt muon pairs in proton-proton collisions at $\sqrt{s}=13$ TeV

Hayrapetyan, Aram et al.

[JHEP 03 \(2026\) 189](#) | [arXiv:2511.11888](#) | [DOI:10.1007/JHEP03\(2026\)189](#) | 2025-11-14

68. Search for a new neutral gauge boson produced in association with one or two b jets and decaying into a pair of muons in proton-proton collisions at $\sqrt{s} = 13$ TeV

Chekhovsky, Vladimir et al.

[arXiv:2511.11853](#) | 2025-11-14

69. Search for long-lived particles using displaced vertices with low-momentum tracks in proton-proton collisions at $\sqrt{s} = 13$ TeV

Hayrapetyan, Aram et al.

[arXiv:2511.08212](#) | 2025-11-11

70. Identification of tau leptons using a convolutional neural network with domain adaptation

Hayrapetyan, Aram et al.

[JINST 20 \(2025\) P12032](#) | [arXiv:2511.05468](#) | [DOI:10.1088/1748-0221/20/12/P12032](#) | 2025-11-07

71. Jet fragmentation function and groomed substructure of bottom quark jets in proton-proton collisions at 5.02 TeV

Chekhovsky, Vladimir et al.

[JHEP 04 \(2026\) 147](#) | [arXiv:2511.10666](#) | [DOI:10.1007/JHEP04\(2026\)147](#) | 2025-11-06

72. Inclusive and differential measurements of the $t\bar{t}\gamma$ cross section and the $t\bar{t}\gamma / t\bar{t}$ cross section ratio in proton-proton collisions at $\sqrt{s} = 13$ TeV

Hayrapetyan, Aram et al.

[JHEP 04 \(2026\) 086](#) | [arXiv:2511.01995](#) | [DOI:10.1007/JHEP04\(2026\)086](#) | 2025-11-03

73. Characterizing the initial state and dynamical evolution in XeXe and PbPb collisions using multiparticle cumulants

Hayrapetyan, Aram et al.

[Phys.Lett.B 876 \(2026\) 140359](#) | [arXiv:2510.26766](#) | [DOI:10.1016/j.physletb.2026.140359](#) | 2025-10-30

74. Search for single production of a vector-like T quark decaying to a top quark and a neutral scalar boson in the lepton+jets final state in proton-proton collisions at $\sqrt{s} = 13$ TeV

Hayrapetyan, Aram et al.

[arXiv:2510.25874](#) | 2025-10-29

75. Measurement of the ratio of the $B_c^+ \rightarrow J/\psi \tau^+ \nu_\tau$ and $B_c^+ \rightarrow J/\psi \mu^+ \nu_\mu$ branching fractions using three-prong τ lepton decays

Chekhovsky, Vladimir et al.

[arXiv:2510.21559](#) | 2025-10-24

76. Search for dijet resonances with data scouting in proton-proton collisions at $\sqrt{s}=13$ TeV

Hayrapetyan, Aram et al.

[JHEP 04 \(2026\) 133](#) | [arXiv:2510.21641](#) | [DOI:10.1007/JHEP04\(2026\)133](#) | 2025-10-24

77. Observation of tWZ Production at the CMS Experiment

Hayrapetyan, Aram et al.

[Phys.Rev.Lett. 136 \(2026\) 081802](#) | [arXiv:2510.19080](#) | [DOI:10.1103/rk6w-1pcl](#) | 2025-10-21

78. Search for dark matter production in association with bottom quarks and a lepton pair in proton-proton collisions at $\sqrt{s} = 13$ TeV

Chekhovsky, Vladimir et al.

[arXiv:2510.12396](#) | 2025-10-14

79. Performance of heavy-flavour jet identification in Lorentz-boosted topologies in proton-proton collisions at $\sqrt{s} = 13$ TeV

Hayrapetyan, Aram et al.

[JINST 20 \(2025\) P11006](#) | [arXiv:2510.10228](#) | [DOI:10.1088/1748-0221/20/11/P11006](#) | 2025-10-11

80. Observation of Suppressed Charged-Particle Production in Ultrarelativistic Oxygen-Oxygen Collisions

Hayrapetyan, Aram et al.

[Phys.Rev.Lett. 136 \(2026\) 162301](#) | [arXiv:2510.09864](#) | [DOI:10.1103/89sf-9t1x](#) | 2025-10-10

81. Combination of searches for nonresonant Higgs boson pair production in proton-proton collisions at $\sqrt{s}= 13$ TeV

Hayrapetyan, Aram et al.
[arXiv:2510.07527](#) | 2025-10-08

82. Observation of long-range collective flow in OO and NeNe collisions and implications for nuclear structure studies

Hayrapetyan, Aram et al.
[arXiv:2510.02580](#) | 2025-10-02

83. Wasserstein normalized autoencoder for anomaly detection

Hayrapetyan, Aram et al.
[arXiv:2510.02168](#) | 2025-10-02

84. Vector boson scattering and anomalous quartic couplings in final states with $\ell\nu qq$ or $\ell\ell qq$ plus jets using proton-proton collisions at $\sqrt{s}=13$ TeV

Hayrapetyan, Aram et al.
[JHEP 03 \(2026\) 022](#) | [arXiv:2510.00118](#) | [DOI:10.1007/JHEP03\(2026\)022](#) | 2025-09-30

85. Measurement of the dineutrino system kinematic variables in dileptonic top quark pair production in proton-proton collisions at $\sqrt{s} = 13$ TeV

Chekhovsky, Vladimir et al.
[arXiv:2510.00160](#) | 2025-09-30

86. Simultaneous Probe of the Charm and Bottom Quark Yukawa Couplings Using $t\bar{t}H$ Events

Hayrapetyan, Aram et al.
[Phys.Rev.Lett. 136 \(2026\) 011801](#) | [arXiv:2509.22535](#) | [DOI:10.1103/9nwb-splk](#) | 2025-09-26

87. Search for resonances decaying to an anomalous jet and a Higgs boson in proton-proton collisions at $\sqrt{s}=13$ TeV

Hayrapetyan, Aram et al.
[Eur.Phys.J.C 86 \(2026\) 137](#) | [arXiv:2509.13635](#) | [DOI:10.1140/epjc/s10052-025-15176-9](#) | 2025-09-17

88. Measurements of $t\bar{t}W$ differential cross sections and the leptonic charge asymmetry at $\sqrt{s}=13$ TeV

Hayrapetyan, Aram et al.
[JHEP 03 \(2026\) 083](#) | [arXiv:2509.13512](#) | [DOI:10.1007/JHEP03\(2026\)083](#) | 2025-09-16

89. Improving missing transverse momentum estimation with a deep neural network

Hayrapetyan, Aram et al.
[Phys.Rev.D 113 \(2026\) 072010](#) | [arXiv:2509.12012](#) | [DOI:10.1103/c4z7-tqvc](#) | 2025-09-15

90. Measurement of D^0 Meson Photoproduction in Ultraperipheral Heavy Ion Collisions

Chekhovsky, Vladimir et al.
[Phys.Rev.Lett. 136 \(2026\) 122303](#) | [arXiv:2509.08626](#) | [DOI:10.1103/lckg-sdh9](#) | 2025-09-10

91. Model-independent measurement of the Higgs boson associated production with two jets and decaying to a pair of W bosons in proton-proton collisions at $\sqrt{s}=13$ TeV

Hayrapetyan, Aram et al.
[JHEP 03 \(2026\) 073](#) | [arXiv:2509.07958](#) | [DOI:10.1007/JHEP03\(2026\)073](#) | 2025-09-09

92. Search for a Higgs boson produced in association with a charm quark and decaying to a W boson pair in proton-proton collisions at $\sqrt{s} = 13$ TeV

Hayrapetyan, Aram et al.
[arXiv:2508.14988](#) | 2025-08-20

93. General search for supersymmetric particles in scenarios with compressed mass spectra using proton-proton collisions at $\sqrt{s}=13$ TeV

Chekhovsky, Vladimir et al.

[Phys.Rev.D 112 \(2025\) 112023](#) | [arXiv:2508.13900](#) | [DOI:10.1103/b26z-zmpy](#) | 2025-08-19

94. Search for a new scalar resonance decaying to a Higgs boson and another new scalar particle in the final state with two bottom quarks and two photons in proton-proton collisions at $\sqrt{s}=13$ TeV

Hayrapetyan, Aram et al.

[JHEP 12 \(2025\) 178](#) | [arXiv:2508.11494](#) | [DOI:10.1007/JHEP12\(2025\)178](#) | 2025-08-15

95. Search for charged lepton flavor violating Z and Z' boson decays in proton-proton collisions at $\sqrt{s}=13$ TeV

Hayrapetyan, Aram et al.

[Phys.Rev.D 112 \(2025\) 112011](#) | [arXiv:2508.07512](#) | [DOI:10.1103/8lw4-6lyk](#) | 2025-08-11

96. Search for light pseudoscalar boson pairs produced from Higgs boson decays using the 4τ and $2\mu 2\tau$ final states in proton-proton collisions at $\sqrt{s}=13$ TeV

Chekhovsky, Vladimir et al.

[JHEP 04 \(2026\) 132](#) | [arXiv:2508.06947](#) | [DOI:10.1007/JHEP04\(2026\)132](#) | 2025-08-09

97. Search for b-hadron decays to long-lived particles in the CMS endcap muon detectors

Hayrapetyan, Aram et al.

[Phys.Rev.D 113 \(2026\) 012009](#) | [arXiv:2508.06363](#) | [DOI:10.1103/7ldn-snzn](#) | 2025-08-08

98. First Exclusive Reconstruction of the B^{*+} , B^{*0} , and B_s^{*0} Mesons and Precise Measurement of Their Masses

Hayrapetyan, Aram et al.

[Phys.Rev.Lett. 136 \(2026\) 031902](#) | [arXiv:2508.05820](#) | [DOI:10.1103/njf9-4zfv](#) | 2025-08-07

99. Search for resonant production of pairs of dijet resonances through broad mediators in proton-proton collisions at $\sqrt{s} = 13$ TeV

Hayrapetyan, Aram et al.

[arXiv:2507.17884](#) | 2025-07-23

100. Probing the flavour structure of dimension-6 EFT operators in multilepton final states in proton-proton collisions at $\sqrt{s}=13$ TeV

Hayrapetyan, Aram et al.

[JHEP 03 \(2026\) 140](#) | [arXiv:2507.17498](#) | [DOI:10.1007/JHEP03\(2026\)140](#) | 2025-07-23

101. Exploring small-angle emissions in charm quark jets in proton-proton collisions at $\sqrt{s} = 5.02$ TeV

Chekhovsky, Vladimir et al.

[arXiv:2507.13469](#) | 2025-07-17

102. Evidence of medium response to hard probes using correlations of Z bosons with hadrons in heavy ion collisions

Chekhovsky, Vladimir et al.

[Phys.Lett.B 874 \(2026\) 140120](#) | [arXiv:2507.09307](#) | [DOI:10.1016/j.physletb.2025.140120](#) | 2025-07-12

103. A method for correcting the substructure of multiprong jets using the Lund jet plane

Hayrapetyan, Aram et al.

[JHEP 11 \(2025\) 038](#) | [arXiv:2507.07775](#) | [DOI:10.1007/JHEP11\(2025\)038](#) | 2025-07-10

104. Search for heavy pseudoscalar and scalar bosons decaying to a top quark pair in proton-proton collisions at $\sqrt{s} = 13$ TeV

Hayrapetyan, Aram et al.

[Rept.Prog.Phys. 88 \(2025\) 127801](#) | [arXiv:2507.05119](#) | [DOI:10.1088/1361-6633/ae2207](#) | 2025-07-07

105. Search for the nonresonant and resonant production of a Higgs boson in association with an additional scalar boson in the $\gamma\gamma\tau\tau$ final state in proton-proton collisions at $\sqrt{s} =$

13 TeV

Hayrapetyan, Aram et al.

[arXiv:2506.23012](#) | 2025-06-28

106. Operation and performance of the CMS silicon strip tracker with proton-proton collisions at the CERN LHC

Hayrapetyan, Aram et al.

[JINST 20 \(2025\) P08027](#) | [arXiv:2506.17195](#) | [DOI:10.1088/1748-0221/20/08/P08027](#) | 2025-06-20

107. Search for nonresonant new physics signals in high-mass dilepton events produced in association with b-tagged jets in proton-proton collisions at $\sqrt{s}=13$ TeV

Chekhovsky, Vladimir et al.

[JHEP 02 \(2026\) 141](#) | [arXiv:2506.13565](#) | [DOI:10.1007/JHEP02\(2026\)141](#) | 2025-06-16

108. Search for top squarks in final states with many light-flavor jets and 0, 1, or 2 charged leptons in proton-proton collisions at $\sqrt{s}=13$ TeV

Chekhovsky, Vladimir et al.

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195. Search for a resonance decaying to a W boson and a photon in proton-proton collisions at $\sqrt{s} = 13$ TeV using leptonic W boson decays

Hayrapetyan, Aram et al.

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[Phys.Rev.Lett. 133 \(2024\) 161805](#) | [arXiv:2404.18298](#) | [DOI:10.1103/PhysRevLett.133.161805](#) | 2024-04-28
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[JINST 19 \(2024\) P09012](#) | [arXiv:2404.17377](#) | [DOI:10.1088/1748-0221/19/09/P09012](#) | 2024-04-26
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[JHEP 10 \(2024\) 209](#) | [arXiv:2404.02711](#) | [DOI:10.1007/JHEP10\(2024\)209](#) | 2024-04-03
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[Phys.Rev.D 109 \(2024\) L111101](#) | [arXiv:2403.11352](#) | [DOI:10.1103/PhysRevD.109.L111101](#) | 2024-03-17

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[JHEP 06 \(2024\) 097](#) | [arXiv:2403.10341](#) | [DOI:10.1007/JHEP06\(2024\)097](#) | 2024-03-15

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[JHEP 06 \(2024\) 123](#) | [arXiv:2403.00100](#) | [DOI:10.1007/JHEP06\(2024\)123](#) | 2024-02-29

232. Search for long-lived heavy neutral leptons decaying in the CMS muon detectors in proton-proton collisions at $\sqrt{s}=13\text{TeV}$

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[Phys.Rev.D 110 \(2024\) 012004](#) | [arXiv:2402.18658](#) | [DOI:10.1103/PhysRevD.110.012004](#) | 2024-02-28

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[Phys.Rev.Lett. 132 \(2024\) 241802](#) | [arXiv:2402.18461](#) | [DOI:10.1103/PhysRevLett.132.241802](#) | 2024-02-28

234. Observation of the $\Xi_b^- \rightarrow \psi(2S) \Xi^-$ decay and studies of the $\Xi_b(5945)^0$ baryon in proton-proton collisions at $\sqrt{s}=13\text{TeV}$

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[Phys.Rev.D 110 \(2024\) 012002](#) | [arXiv:2402.17738](#) | [DOI:10.1103/PhysRevD.110.012002](#) | 2024-02-27

235. Search for long-lived particles using displaced vertices and missing transverse momentum in proton-proton collisions at $\sqrt{s}=13\text{TeV}$

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[Phys.Rev.D 109 \(2024\) 112005](#) | [arXiv:2402.15804](#) | [DOI:10.1103/PhysRevD.109.112005](#) | 2024-02-24

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[Comput.Softw.Big Sci. 8 \(2024\) 17](#) | [arXiv:2402.15366](#) | [DOI:10.1007/s41781-024-00124-1](#) | 2024-02-23

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[JHEP 05 \(2024\) 047](#) | [arXiv:2402.14491](#) | [DOI:10.1007/JHEP05\(2024\)047](#) | 2024-02-22

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[Phys.Rev.D 110 \(2024\) 052004](#) | [arXiv:2402.13808](#) | [DOI:10.1103/PhysRevD.110.052004](#) | 2024-02-21

240. Search for exotic decays of the Higgs boson to a pair of pseudoscalars in the $\mu\mu b\bar{b}$ and $\tau\tau b\bar{b}$ final states

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[Eur.Phys.J.C 84 \(2024\) 493](#) | [arXiv:2402.13358](#) | [DOI:10.1140/epjc/s10052-024-12727-4](#) | 2024-02-20

241. Search for a scalar or pseudoscalar dilepton resonance produced in association with a massive vector boson or top quark-antiquark pair in multilepton events at $\sqrt{s}=13\text{TeV}$

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[Phys.Rev.D 110 \(2024\) 012013](#) | [arXiv:2402.11098](#) | [DOI:10.1103/PhysRevD.110.012013](#) | 2024-02-16

242. Search for Fractionally Charged Particles in Proton-Proton Collisions at $\sqrt{s}=13\text{TeV}$

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[Phys.Rev.Lett. 134 \(2025\) 131802](#) | [arXiv:2402.09932](#) | [DOI:10.1103/PhysRevLett.134.131802](#) | 2024-02-15

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[Phys.Rev.Lett. 132 \(2024\) 261902](#) | [arXiv:2402.08713](#) | [DOI:10.1103/PhysRevLett.132.261902](#) | 2024-02-13

244. Differential cross section measurements for the production of top quark pairs and of additional jets using dilepton events from pp collisions at $\sqrt{s} = 13 \text{ TeV}$

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[JHEP 02 \(2025\) 064](#) | [arXiv:2402.08486](#) | [DOI:10.1007/JHEP02\(2025\)064](#) | 2024-02-13

245. Search for pair production of scalar and vector leptoquarks decaying to muons and bottom quarks in proton-proton collisions at $\sqrt{s}=13\text{TeV}$

Hayrapetyan, Aram et al.

[Phys.Rev.D 109 \(2024\) 112003](#) | [arXiv:2402.08668](#) | [DOI:10.1103/PhysRevD.109.112003](#) | 2024-02-09

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[Phys.Rev.D 110 \(2024\) 032007](#) | [arXiv:2402.01898](#) | [DOI:10.1103/PhysRevD.110.032007](#) | 2024-02-02

247. Combined search for electroweak production of winos, binos, higgsinos, and sleptons in proton-proton collisions at $\sqrt{s}=13\text{TeV}$

Hayrapetyan, Aram et al.

[Phys.Rev.D 109 \(2024\) 112001](#) | [arXiv:2402.01888](#) | [DOI:10.1103/PhysRevD.109.112001](#) | 2024-02-02

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[Phys.Rev.D 109 \(2024\) 112013](#) | [arXiv:2401.14494](#) | [DOI:10.1103/PhysRevD.109.112013](#) | 2024-01-25

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[JHEP 01 \(2025\) 011](#) | [arXiv:2401.11355](#) | [DOI:10.1007/JHEP01\(2025\)011](#) | 2024-01-20

251. Test of lepton flavor universality in $B^{\pm} \rightarrow K^{\pm} \mu^+ \mu^-$ and $B^{\pm} \rightarrow K^{\pm} e^+ e^-$ decays in proton-proton collisions at $\sqrt{s} = 13 \text{ TeV}$

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[Rept.Prog.Phys. 87 \(2024\) 077802](#) | [arXiv:2401.07090](#) | [DOI:10.1088/1361-6633/ad4e65](#) | 2024-01-13

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[Rept.Prog.Phys. 87 \(2024\) 077801](#) | [arXiv:2401.06896](#) | [DOI:10.1088/1361-6633/ad4b9b](#) | 2024-01-12

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[Nature Commun. 16 \(2025\) 7990](#) | [arXiv:2312.17092](#) | [DOI:10.1038/s41467-025-56200-6](#) | 2023-12-28

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[Eur.Phys.J.C 85 \(2025\) 72](#) | [arXiv:2312.16669](#) | [DOI:10.1140/epjc/s10052-024-13606-8](#) | 2023-12-27

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[JHEP 05 \(2024\) 116](#) | [arXiv:2312.16343](#) | [DOI:10.1007/JHEP05\(2024\)116](#) | 2023-12-26

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[JHEP 03 \(2024\) 105](#) | [arXiv:2312.07484](#) | [DOI:10.1007/JHEP03\(2024\)105](#) | 2023-12-12

259. Search for flavor changing neutral current interactions of the top quark in final states with a photon and additional jets in proton-proton collisions at $\sqrt{s}=13\text{TeV}$

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[Phys.Rev.D 109 \(2024\) 072004](#) | [arXiv:2312.08229](#) | [DOI:10.1103/PhysRevD.109.072004](#) | 2023-12-11

260. Measurement of simplified template cross sections of the Higgs boson produced in association with W or Z bosons in the $H\rightarrow b\bar{b}$ decay channel in proton-proton collisions at $\sqrt{s}=13\text{TeV}$

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[Phys.Rev.D 109 \(2024\) 092011](#) | [arXiv:2312.07562](#) | [DOI:10.1103/PhysRevD.109.092011](#) | 2023-12-09

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[Phys.Lett.B 853 \(2024\) 138633](#) | [arXiv:2312.02371](#) | [DOI:10.1016/j.physletb.2024.138633](#) | 2023-12-04

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[JHEP 2024 \(2024\) 106](#) | [arXiv:2311.11370](#) | [DOI:10.1007/JHEP02\(2024\)106](#) | 2023-11-19

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[Phys.Lett.B 850 \(2024\) 138478](#) | [arXiv:2311.03261](#) | [DOI:10.1016/j.physletb.2024.138478](#) | 2023-11-06

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[Phys.Rev.D 110 \(2024\) 012010](#) | [arXiv:2311.02725](#) | [DOI:10.1103/PhysRevD.110.012010](#) | 2023-11-05

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[Phys.Lett.B 852 \(2024\) 138582](#) | [arXiv:2311.00130](#) | [DOI:10.1016/j.physletb.2024.138582](#) | 2023-10-31

267. Search for W' bosons decaying to a top and a bottom quark in leptonic final states in proton-proton collisions at $\sqrt{s} = 13\text{ TeV}$

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[JHEP 05 \(2024\) 046](#) | [arXiv:2310.19893](#) | [DOI:10.1007/JHEP05\(2024\)046](#) | 2023-10-30

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[Phys.Rev.Lett. 133 \(2024\) 011801](#) | [arXiv:2310.14023](#) | [DOI:10.1103/PhysRevLett.133.011801](#) | 2023-10-21

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Tumasyan, Armen et al.

[JHEP 06 \(2024\) 187](#) | [arXiv:2310.11231](#) | [DOI:10.1007/JHEP06\(2024\)187](#) | 2023-10-17

270. Observation of $WW\gamma$ production and search for $H\gamma$ production in proton-proton collisions at $\sqrt{s} = 13$ TeV

Hayrapetyan, Aram et al.

[Phys.Rev.Lett. 132 \(2024\) 121901](#) | [arXiv:2310.05164](#) | [DOI:10.1103/PhysRevLett.132.121901](#) | 2023-10-08

271. Muon identification using multivariate techniques in the CMS experiment in proton-proton collisions at $\sqrt{s} = 13$ TeV

Hayrapetyan, Aram et al.

[JINST 19 \(2024\) P02031](#) | [arXiv:2310.03844](#) | [DOI:10.1088/1748-0221/19/02/P02031](#) | 2023-10-05

272. Study of azimuthal anisotropy of $\phi(1S)$ mesons in pPb collisions at $\sqrt{s_{NN}} = 8.16$ TeV

Tumasyan, Armen et al.

[Phys.Lett.B 850 \(2024\) 138518](#) | [arXiv:2310.03233](#) | [DOI:10.1016/j.physletb.2024.138518](#) | 2023-10-04

273. Search for stealth supersymmetry in final states with two photons, jets, and low missing transverse momentum in proton-proton collisions at $\sqrt{s}=13$ TeV

Hayrapetyan, Aram et al.

[Phys.Rev.D 109 \(2024\) 112009](#) | [arXiv:2310.03154](#) | [DOI:10.1103/PhysRevD.109.112009](#) | 2023-10-04

274. Search for a new resonance decaying into two spin-0 bosons in a final state with two photons and two bottom quarks in proton-proton collisions at $\sqrt{s} = 13$ TeV

Tumasyan, Armen et al.

[JHEP 05 \(2024\) 316](#) | [arXiv:2310.01643](#) | [DOI:10.1007/JHEP05\(2024\)316](#) | 2023-10-02

275. Search for supersymmetry in final states with disappearing tracks in proton-proton collisions at $\sqrt{s}=13$ TeV

Hayrapetyan, Aram et al.

[Phys.Rev.D 109 \(2024\) 072007](#) | [arXiv:2309.16823](#) | [DOI:10.1103/PhysRevD.109.072007](#) | 2023-09-28

276. Search for direct production of GeV-scale resonances decaying to a pair of muons in proton-proton collisions at $\sqrt{s} = 13$ TeV

Hayrapetyan, Aram et al.

[JHEP 12 \(2023\) 070](#) | [arXiv:2309.16003](#) | [DOI:10.1007/JHEP12\(2023\)070](#) | 2023-09-27

277. Inclusive and differential cross section measurements of $t\bar{t}b\bar{b}$ production in the lepton+jets channel at $\sqrt{s} = 13$ TeV

Hayrapetyan, Aram et al.

[JHEP 05 \(2024\) 042](#) | [arXiv:2309.14442](#) | [DOI:10.1007/JHEP05\(2024\)042](#) | 2023-09-25

278. Measurement of the τ lepton polarization in Z boson decays in proton-proton collisions at $\sqrt{s} = 13$ TeV

Hayrapetyan, Aram et al.

[JHEP 01 \(2024\) 101](#) | [arXiv:2309.12408](#) | [DOI:10.1007/JHEP01\(2024\)101](#) | 2023-09-21

279. Development of the CMS detector for the CERN LHC Run 3

Hayrapetyan, Aram et al.

[JINST 19 \(2024\) P05064](#) | [arXiv:2309.05466](#) | [DOI:10.1088/1748-0221/19/05/P05064](#) | 2023-09-11

280. Evidence for the Higgs Boson Decay to a Z Boson and a Photon at the LHC

Aad, Georges et al.

[Phys.Rev.Lett. 132 \(2024\) 021803](#) | [arXiv:2309.03501](#) | [DOI:10.1103/PhysRevLett.132.021803](#) | 2023-09-07

281. Luminosity determination using Z boson production at the CMS experiment

Hayrapetyan, Aram et al.

[Eur.Phys.J.C 84 \(2024\) 26](#) | [arXiv:2309.01008](#) | [DOI:10.1140/epjc/s10052-023-12268-2](#) | 2023-09-02

282. Search for a third-generation leptoquark coupled to a τ lepton and a b quark through single, pair, and nonresonant production in proton-proton collisions at $\sqrt{s} = 13$ TeV

Hayrapetyan, Aram et al.

[JHEP 05 \(2024\) 311](#) | [arXiv:2308.07826](#) | [DOI:10.1007/JHEP05\(2024\)311](#) | 2023-08-15

283. Search for Scalar Leptoquarks Produced via τ -Lepton-Quark Scattering in pp Collisions at $\sqrt{s}=13\text{TeV}$

Hayrapetyan, Aram et al.

[Phys.Rev.Lett. 132 \(2024\) 061801](#) | [arXiv:2308.06143](#) | [DOI:10.1103/PhysRevLett.132.061801](#) | 2023-08-11

284. Measurement of the production cross section for a W boson in association with a charm quark in proton-proton collisions at $\sqrt{s} = 13 \text{ TeV}$

Tumasyan, Armen et al.

[Eur.Phys.J.C 84 \(2024\) 27](#) | [arXiv:2308.02285](#) | [DOI:10.1140/epjc/s10052-023-12258-4](#) | 2023-08-04

285. Measurement of the Higgs boson production via vector boson fusion and its decay into bottom quarks in proton-proton collisions at $\sqrt{s} = 13 \text{ TeV}$

Hayrapetyan, Aram et al.

[JHEP 01 \(2024\) 173](#) | [arXiv:2308.01253](#) | [DOI:10.1007/JHEP01\(2024\)173](#) | 2023-08-02

286. Search for new physics in multijet events with at least one photon and large missing transverse momentum in proton-proton collisions at 13 TeV

Hayrapetyan, Aram et al.

[JHEP 10 \(2023\) 046](#) | [arXiv:2307.16216](#) | [DOI:10.1007/JHEP10\(2023\)046](#) | 2023-07-30

287. Search for physics beyond the standard model in top quark production with additional leptons in the context of effective field theory

Hayrapetyan, Aram et al.

[JHEP 12 \(2023\) 068](#) | [arXiv:2307.15761](#) | [DOI:10.1007/JHEP12\(2023\)068](#) | 2023-07-28

288. Study of charm hadronization with prompt Λ_c^+ baryons in proton-proton and lead-lead collisions at $\sqrt{s_{NN}} = 5.02 \text{ TeV}$

Tumasyan, Armen et al.

[JHEP 01 \(2024\) 128](#) | [arXiv:2307.11186](#) | [DOI:10.1007/JHEP01\(2024\)128](#) | 2023-07-20

289. Multiplicity and transverse momentum dependence of charge-balance functions in pPb and PbPb collisions at LHC energies

Tumasyan, Armen et al.

[JHEP 08 \(2024\) 148](#) | [arXiv:2307.11185](#) | [DOI:10.1007/JHEP08\(2024\)148](#) | 2023-07-20

290. Search for a high-mass dimuon resonance produced in association with b quark jets at $\sqrt{s} = 13 \text{ TeV}$

Hayrapetyan, Aram et al.

[JHEP 10 \(2023\) 043](#) | [arXiv:2307.08708](#) | [DOI:10.1007/JHEP10\(2023\)043](#) | 2023-07-14

291. Search for Z' bosons decaying to pairs of heavy Majorana neutrinos in proton-proton collisions at $\sqrt{s} = 13 \text{ TeV}$

Tumasyan, Armen et al.

[JHEP 11 \(2023\) 181](#) | [arXiv:2307.06959](#) | [DOI:10.1007/JHEP11\(2023\)181](#) | 2023-07-12

292. Two-particle Bose-Einstein correlations and their Lévy parameters in PbPb collisions at $\sqrt{s_{NN}}=5.02 \text{ TeV}$

Tumasyan, Armen et al.

[Phys.Rev.C 109 \(2024\) 024914](#) | [arXiv:2306.11574](#) | [DOI:10.1103/PhysRevC.109.024914](#) | 2023-06-20

293. Performance of the local reconstruction algorithms for the CMS hadron calorimeter with Run 2 data

Tumasyan, Armen et al.

[JINST 18 \(2023\) P11017](#) | [arXiv:2306.10355](#) | [DOI:10.1088/1748-0221/18/11/P11017](#) | 2023-06-17

294. New Structures in the $J/\psi/\psi$ Mass Spectrum in Proton-Proton Collisions at $\sqrt{s}=13\text{TeV}$
Hayrapetyan, Aram et al.
[Phys.Rev.Lett. 132 \(2024\) 111901](#) | [arXiv:2306.07164](#) | [DOI:10.1103/PhysRevLett.132.111901](#) | 2023-06-12
295. Search for the lepton-flavor violating decay of the Higgs boson and additional Higgs bosons in the $e\mu$ final state in proton-proton collisions at $\sqrt{s} = 13 \text{ TeV}$
Hayrapetyan, Aram et al.
[Phys.Rev.D 108 \(2023\) 072004](#) | [arXiv:2305.18106](#) | [DOI:10.1103/PhysRevD.108.072004](#) | 2023-05-29
296. Measurements of the azimuthal anisotropy of prompt and nonprompt charmonia in PbPb collisions at $\sqrt{s_{\text{NN}}} = 5.02 \text{ TeV}$
Tumasyan, Armen et al.
[JHEP 10 \(2023\) 115](#) | [arXiv:2305.16928](#) | [DOI:10.1007/JHEP10\(2023\)115](#) | 2023-05-26
297. Observation of four top quark production in proton-proton collisions at $\sqrt{s}=13 \text{ TeV}$
Hayrapetyan, Aram et al.
[Phys.Lett.B 847 \(2023\) 138290](#) | [arXiv:2305.13439](#) | [DOI:10.1016/j.physletb.2023.138290](#) | 2023-05-22
298. Search for Inelastic Dark Matter in Events with Two Displaced Muons and Missing Transverse Momentum in Proton-Proton Collisions at $\sqrt{s}=13\text{TeV}$
Hayrapetyan, Aram et al.
[Phys.Rev.Lett. 132 \(2024\) 041802](#) | [arXiv:2305.11649](#) | [DOI:10.1103/PhysRevLett.132.041802](#) | 2023-05-19
299. Search for resonances in events with photon and jet final states in proton-proton collisions at $\sqrt{s} = 13 \text{ TeV}$
Tumasyan, Armen et al.
[JHEP 12 \(2023\) 189](#) | [arXiv:2305.07998](#) | [DOI:10.1007/JHEP12\(2023\)189](#) | 2023-05-13
300. Measurements of inclusive and differential cross sections for the Higgs boson production and decay to four-leptons in proton-proton collisions at $\sqrt{s} = 13 \text{ TeV}$
Hayrapetyan, Aram et al.
[JHEP 08 \(2023\) 040](#) | [arXiv:2305.07532](#) | [DOI:10.1007/JHEP08\(2023\)040](#) | 2023-05-12
301. Observation of the rare decay of the η meson to four muons
Hayrapetyan, Aram et al.
[Phys.Rev.Lett. 131 \(2023\) 091903](#) | [arXiv:2305.04904](#) | [DOI:10.1103/PhysRevLett.131.091903](#) | 2023-05-08
302. Search for top squark pair production in a final state with at least one hadronically decaying tau lepton in proton-proton collisions at $\sqrt{s} = 13 \text{ TeV}$
Tumasyan, Armen et al.
[JHEP 07 \(2023\) 110](#) | [arXiv:2304.07174](#) | [DOI:10.1007/JHEP07\(2023\)110](#) | 2023-04-14
303. Probing Small Bjorken- x Nuclear Gluonic Structure via Coherent J/ψ Photoproduction in Ultraperipheral Pb-Pb Collisions at $\sqrt{s_{\text{NN}}}=5.02\text{TeV}$
Tumasyan, Armen et al.
[Phys.Rev.Lett. 131 \(2023\) 262301](#) | [arXiv:2303.16984](#) | [DOI:10.1103/PhysRevLett.131.262301](#) | 2023-03-29
304. Observation of the $\chi(3S)$ Meson and Suppression of χ States in Pb-Pb Collisions at $\sqrt{s_{\text{NN}}}=5.02\text{TeV}$
Tumasyan, Armen et al.
[Phys.Rev.Lett. 133 \(2024\) 022302](#) | [arXiv:2303.17026](#) | [DOI:10.1103/PhysRevLett.133.022302](#) | 2023-03-29
305. First measurement of the top quark pair production cross section in proton-proton collisions at $\sqrt{s} = 13.6 \text{ TeV}$
Tumasyan, Armen et al.
[JHEP 08 \(2023\) 204](#) | [arXiv:2303.10680](#) | [DOI:10.1007/JHEP08\(2023\)204](#) | 2023-03-19

306. A search for new physics in central exclusive production using the missing mass technique with the CMS detector and the CMS-TOTEM precision proton spectrometer
Tumasyan, Armen et al.
[Eur.Phys.J.C 83 \(2023\) 827](#) | [arXiv:2303.04596](#) | [DOI:10.1140/epjc/s10052-023-11687-5](#) | 2023-03-08
307. Evidence for Four-Top Quark Production in Proton-Proton Collisions at $\sqrt{s}=13$ TeV
Tumasyan, Armen et al.
[Phys.Lett.B 844 \(2023\) 138076](#) | [arXiv:2303.03864](#) | [DOI:10.1016/j.physletb.2023.138076](#) | 2023-03-07
308. A search for decays of the Higgs boson to invisible particles in events with a top-antitop quark pair or a vector boson in proton-proton collisions at $\sqrt{s} = 13$ TeV
Tumasyan, Armen et al.
[Eur.Phys.J.C 83 \(2023\) 933](#) | [arXiv:2303.01214](#) | [DOI:10.1140/epjc/s10052-023-11952-7](#) | 2023-03-02
309. Search for dark matter particles in W^+W^- events with transverse momentum imbalance in proton-proton collisions at $\sqrt{s} = 13$ TeV
Hayrapetyan, Aram et al.
[JHEP 03 \(2024\) 134](#) | [arXiv:2310.12229](#) | [DOI:10.1007/JHEP03\(2024\)134](#) | 2023-03
310. Search for a vector-like quark $T' \rightarrow tH$ via the diphoton decay mode of the Higgs boson in proton-proton collisions at $\sqrt{s} = 13$ TeV
Tumasyan, Armen et al.
[JHEP 09 \(2023\) 057](#) | [arXiv:2302.12802](#) | [DOI:10.1007/JHEP09\(2023\)057](#) | 2023-02-24
311. Measurement of the top quark mass using a profile likelihood approach with the lepton + jets final states in proton-proton collisions at $\sqrt{s}=13$ TeV
Tumasyan, Armen et al.
[Eur.Phys.J.C 83 \(2023\) 963](#) | [arXiv:2302.01967](#) | [DOI:10.1140/epjc/s10052-023-12050-4](#) | 2023-02-03
312. Search for top squarks in the four-body decay mode with single lepton final states in proton-proton collisions at $\sqrt{s} = 13$ TeV
Tumasyan, Armen et al.
[JHEP 06 \(2023\) 060](#) | [arXiv:2301.08096](#) | [DOI:10.1007/JHEP06\(2023\)060](#) | 2023-01-19
313. First Measurement of the Forward Rapidity Gap Distribution in pPb Collisions at $\sqrt{s_{NN}}=8.16$ TeV
Tumasyan, Armen et al.
[Phys.Rev.D 108 \(2023\) 092004](#) | [arXiv:2301.07630](#) | [DOI:10.1103/PhysRevD.108.092004](#) | 2023-01-18
314. K_S^0 and $\Lambda(\Lambda^0)$ two-particle femtoscopic correlations in PbPb collisions at $\sqrt{s_{NN}}=5.02$ TeV
Tumasyan, Armen et al.
[Phys.Lett.B 857 \(2024\) 138936](#) | [arXiv:2301.05290](#) | [DOI:10.1016/j.physletb.2024.138936](#) | 2023-01-12
315. Search for new physics in the τ lepton plus missing transverse momentum final state in proton-proton collisions at $\sqrt{s} = 13$ TeV
Tumasyan, A. et al.
[JHEP 09 \(2023\) 051](#) | [arXiv:2212.12604](#) | [DOI:10.1007/JHEP09\(2023\)051](#) | 2022-12-23
316. Measurement of the electroweak production of $W\gamma$ in association with two jets in proton-proton collisions at $\sqrt{s} = 13$ TeV
Tumasyan, Armen et al.
[Phys.Rev.D 108 \(2023\) 032017](#) | [arXiv:2212.12592](#) | [DOI:10.1103/PhysRevD.108.032017](#) | 2022-12-23
317. Measurement of the $B^0_S \rightarrow \mu^+ \mu^-$ decay properties and search for the $B^0 \rightarrow \mu^+ \mu^-$ decay in proton-proton collisions at $\sqrt{s} = 13$ TeV
Tumasyan, Armen et al.

[Phys.Lett.B 842 \(2023\) 137955](#) | [arXiv:2212.10311](#) | [DOI:10.1016/j.physletb.2023.137955](#) | 2022-12-20

318. Search for long-lived particles using out-of-time trackless jets in proton-proton collisions at $\sqrt{s} = 13$ TeV

Tumasyan, Armen et al.

[JHEP 07 \(2023\) 210](#) | [arXiv:2212.06695](#) | [DOI:10.1007/JHEP07\(2023\)210](#) | 2022-12-13

319. Measurement of the dependence of the hadron production fraction ratio f_s / f_u and f_d / f_u on B meson kinematic variables in proton-proton collisions at $\sqrt{s} = 13$ TeV

Tumasyan, Armen et al.

[Phys.Rev.Lett. 131 \(2023\) 121901](#) | [arXiv:2212.02309](#) | [DOI:10.1103/PhysRevLett.131.121901](#) | 2022-12-05

320. Measurements of azimuthal anisotropy of nonprompt D0 mesons in PbPb collisions at $\sqrt{s_{NN}}=5.02$ TeV

Tumasyan, Armen et al.

[Phys.Lett.B 850 \(2024\) 138389](#) | [arXiv:2212.01636](#) | [DOI:10.1016/j.physletb.2023.138389](#) | 2022-12-03

321. Search for high-mass exclusive $\gamma\gamma \rightarrow WW$ and $\gamma\gamma \rightarrow ZZ$ production in proton-proton collisions at $\sqrt{s} = 13$ TeV

Tumasyan, Armen et al.

[JHEP 07 \(2023\) 229](#) | [arXiv:2211.16320](#) | [DOI:10.1007/JHEP07\(2023\)229](#) | 2022-11-29

322. Search for Higgs Boson and Observation of Z Boson through their Decay into a Charm Quark-Antiquark Pair in Boosted Topologies in Proton-Proton Collisions at $\sqrt{s}=13$ TeV

Tumasyan, Armen et al.

[Phys.Rev.Lett. 131 \(2023\) 041801](#) | [arXiv:2211.14181](#) | [DOI:10.1103/PhysRevLett.131.041801](#) | 2022-11-25

323. Search for supersymmetry in final states with a single electron or muon using angular correlations and heavy-object identification in proton-proton collisions at $\sqrt{s} = 13$ TeV

Tumasyan, Armen et al.

[JHEP 09 \(2023\) 149](#) | [arXiv:2211.08476](#) | [DOI:10.1007/JHEP09\(2023\)149](#) | 2022-11-15

324. Measurement of the differential tt production cross section as a function of the jet mass and extraction of the top quark mass in hadronic decays of boosted top quarks

Tumasyan, Armen et al.

[Eur.Phys.J.C 83 \(2023\) 560](#) | [arXiv:2211.01456](#) | [DOI:10.1140/epjc/s10052-023-11587-8](#) | 2022-11-02

325. Azimuthal correlations in Z +jets events in proton-proton collisions at $\sqrt{s} = 13$ TeV

Tumasyan, Armen et al.

[Eur.Phys.J.C 83 \(2023\) 722](#) | [arXiv:2210.16139](#) | [DOI:10.1140/epjc/s10052-023-11833-z](#) | 2022-10-28

326. Measurements of jet multiplicity and jet transverse momentum in multijet events in proton-proton collisions at $\sqrt{s}=13$ TeV

Tumasyan, Armen et al.

[Eur.Phys.J.C 83 \(2023\) 742](#) | [arXiv:2210.13557](#) | [DOI:10.1140/epjc/s10052-023-11753-y](#) | 2022-10-24

327. Search for medium effects using jets from bottom quarks in PbPb collisions at $\sqrt{s_{NN}}=5.02$ TeV

Tumasyan, Armen et al.

[Phys.Lett.B 844 \(2023\) 137849](#) | [arXiv:2210.08547](#) | [DOI:10.1016/j.physletb.2023.137849](#) | 2022-10-16

328. Azimuthal anisotropy of dijet events in PbPb collisions at $\sqrt{s_{NN}} = 5.02$ TeV

Tumasyan, Armen et al.

[JHEP 07 \(2023\) 139](#) | [arXiv:2210.08325](#) | [DOI:10.1007/JHEP07\(2023\)139](#) | 2022-10-15

329. Proton reconstruction with the CMS-TOTEM Precision Proton Spectrometer

Tumasyan, Armen et al.

[JINST 18 \(2023\) P09009](#) | [arXiv:2210.05854](#) | [DOI:10.1088/1748-0221/18/09/P09009](#) | 2022-10-11

330. Search for a heavy composite Majorana neutrino in events with dilepton signatures from proton-proton collisions at $\sqrt{s}=13\text{TeV}$

Tumasyan, Armen et al.

[Phys.Lett.B 843 \(2023\) 137803](#) | [arXiv:2210.03082](#) | [DOI:10.1016/j.physletb.2023.137803](#) | 2022-10-06

331. Search for new heavy resonances decaying to WW, WZ, ZZ, WH, or ZH boson pairs in the all-jets final state in proton-proton collisions at $\sqrt{s}=13\text{ TeV}$

Tumasyan, Armen et al.

[Phys.Lett.B 844 \(2023\) 137813](#) | [arXiv:2210.00043](#) | [DOI:10.1016/j.physletb.2023.137813](#) | 2022-09-30

332. Search for pair production of vector-like quarks in leptonic final states in proton-proton collisions at $\sqrt{s} = 13\text{ TeV}$

Tumasyan, Armen et al.

[JHEP 07 \(2023\) 020](#) | [arXiv:2209.07327](#) | [DOI:10.1007/JHEP07\(2023\)020](#) | 2022-09-15

333. Search for exotic Higgs boson decays $H \rightarrow \mathcal{A}\mathcal{A} \rightarrow 4\gamma$ with events containing two merged diphotons in proton-proton collisions at $\sqrt{s} = 13\text{ TeV}$

Tumasyan, Armen et al.

[Phys.Rev.Lett. 131 \(2023\) 101801](#) | [arXiv:2209.06197](#) | [DOI:10.1103/PhysRevLett.131.101801](#) | 2022-09-13

334. Search for new physics using effective field theory in 13 TeV pp collision events that contain a top quark pair and a boosted Z or Higgs boson

Tumasyan, Armen et al.

[Phys.Rev.D 108 \(2023\) 032008](#) | [arXiv:2208.12837](#) | [DOI:10.1103/PhysRevD.108.032008](#) | 2022-08-26

335. Measurement of the Higgs boson inclusive and differential fiducial production cross sections in the diphoton decay channel with pp collisions at $\sqrt{s} = 13\text{ TeV}$

Tumasyan, Armen et al.

[JHEP 07 \(2023\) 091](#) | [arXiv:2208.12279](#) | [DOI:10.1007/JHEP07\(2023\)091](#) | 2022-08-25

336. Search for pair-produced vector-like leptons in final states with third-generation leptons and at least three b quark jets in proton-proton collisions at $\sqrt{s}=13\text{ TeV}$

Tumasyan, Armen et al.

[Phys.Lett.B 846 \(2023\) 137713](#) | [arXiv:2208.09700](#) | [DOI:10.1016/j.physletb.2023.137713](#) | 2022-08-20

337. Measurement of the cross section of top quark-antiquark pair production in association with a W boson in proton-proton collisions at $\sqrt{s} = 13\text{ TeV}$

Tumasyan, Armen et al.

[JHEP 07 \(2023\) 219](#) | [arXiv:2208.06485](#) | [DOI:10.1007/JHEP07\(2023\)219](#) | 2022-08-12

338. Search for CP violation in ttH and tH production in multilepton channels in proton-proton collisions at $\sqrt{s} = 13\text{ TeV}$

Tumasyan, Armen et al.

[JHEP 07 \(2023\) 092](#) | [arXiv:2208.02686](#) | [DOI:10.1007/JHEP07\(2023\)092](#) | 2022-08-04

339. Measurement of the $t\bar{t}$ charge asymmetry in events with highly Lorentz-boosted top quarks in pp collisions at $\sqrt{s}=13\text{ TeV}$

Tumasyan, Armen et al.

[Phys.Lett.B 846 \(2023\) 137703](#) | [arXiv:2208.02751](#) | [DOI:10.1016/j.physletb.2023.137703](#) | 2022-08-04

340. Searches for additional Higgs bosons and for vector leptoquarks in $\tau\tau$ final states in proton-proton collisions at $\sqrt{s} = 13\text{ TeV}$

Tumasyan, Armen et al.

[JHEP 07 \(2023\) 073](#) | [arXiv:2208.02717](#) | [DOI:10.1007/JHEP07\(2023\)073](#) | 2022-08-04

341. Search for the exotic decay of the Higgs boson into two light pseudoscalars with four photons in the final state in proton-proton collisions at $\sqrt{s} = 13$ TeV

Tumasyan, Armen et al.

[JHEP 07 \(2023\) 148](#) | [arXiv:2208.01469](#) | [DOI:10.1007/JHEP07\(2023\)148](#) | 2022-08-02

342. Measurement of inclusive and differential cross sections for single top quark production in association with a W boson in proton-proton collisions at $\sqrt{s} = 13$ TeV

Tumasyan, Armen et al.

[JHEP 07 \(2023\) 046](#) | [arXiv:2208.00924](#) | [DOI:10.1007/JHEP07\(2023\)046](#) | 2022-08-01

343. Search for the Higgs boson decay to a pair of electrons in proton-proton collisions at $\sqrt{s}=13$ TeV

Tumasyan, Armen et al.

[Phys.Lett.B 846 \(2023\) 137783](#) | [arXiv:2208.00265](#) | [DOI:10.1016/j.physletb.2023.137783](#) | 2022-07-30

344. Search for new physics in the lepton plus missing transverse momentum final state in proton-proton collisions at $\sqrt{s} = 13$ TeV

Tumasyan, Armen et al.

[JHEP 07 \(2022\) 067](#) | [arXiv:2202.06075](#) | [DOI:10.1007/JHEP07\(2022\)067](#) | 2022-07-11

345. Measurement of the top quark pole mass using $t\bar{t}$ +jet events in the dilepton final state in proton-proton collisions at $\sqrt{s} = 13$ TeV

Tumasyan, Armen et al.

[JHEP 07 \(2023\) 077](#) | [arXiv:2207.02270](#) | [DOI:10.1007/JHEP07\(2023\)077](#) | 2022-07-05

346. Search for direct pair production of supersymmetric partners of τ leptons in the final state with two hadronically decaying τ leptons and missing transverse momentum in proton-proton collisions at $\sqrt{s} = 13$ TeV

Tumasyan, Armen et al.

[Phys.Rev.D 108 \(2023\) 012011](#) | [arXiv:2207.02254](#) | [DOI:10.1103/PhysRevD.108.012011](#) | 2022-07-05

347. Search for a charged Higgs boson decaying into a heavy neutral Higgs boson and a W boson in proton-proton collisions at $\sqrt{s} = 13$ TeV

Tumasyan, Armen et al.

[JHEP 09 \(2023\) 032](#) | [arXiv:2207.01046](#) | [DOI:10.1007/JHEP09\(2023\)032](#) | 2022-07-03

348. Search for Higgs boson pairs decaying to WW^*WW^* , $WW^*\tau\tau$, and $\tau\tau\tau\tau$ in proton-proton collisions at $\sqrt{s} = 13$ TeV

Tumasyan, Armen et al.

[JHEP 07 \(2023\) 095](#) | [arXiv:2206.10268](#) | [DOI:10.1007/JHEP07\(2023\)095](#) | 2022-06-21

349. Search for nonresonant Higgs boson pair production in the four leptons plus two b jets final state in proton-proton collisions at $\sqrt{s} = 13$ TeV

Tumasyan, Armen et al.

[JHEP 06 \(2023\) 130](#) | [arXiv:2206.10657](#) | [DOI:10.1007/JHEP06\(2023\)130](#) | 2022-06-21

350. Search for nonresonant Higgs boson pair production in final state with two bottom quarks and two tau leptons in proton-proton collisions at $\sqrt{s}=13$ TeV

Tumasyan, Armen et al.

[Phys.Lett.B 842 \(2023\) 137531](#) | [arXiv:2206.09401](#) | [DOI:10.1016/j.physletb.2022.137531](#) | 2022-06-19

351. Measurements of the Higgs boson production cross section and couplings in the W boson pair decay channel in proton-proton collisions at $\sqrt{s}=13$ TeV

Tumasyan, Armen et al.

[Eur.Phys.J.C 83 \(2023\) 667](#) | [arXiv:2206.09466](#) | [DOI:10.1140/epjc/s10052-023-11632-6](#) | 2022-06-19

352. Probing Heavy Majorana Neutrinos and the Weinberg Operator through Vector Boson Fusion Processes in Proton-Proton Collisions at $\sqrt{s}=13\text{TeV}$

Tumasyan, Armen et al.

[Phys.Rev.Lett. 131 \(2023\) 011803](#) | [arXiv:2206.08956](#) | [DOI:10.1103/PhysRevLett.131.011803](#) | 2022-06-17

353. Precision measurement of the Z boson invisible width in pp collisions at $\sqrt{s}=13\text{ TeV}$

Tumasyan, Armen et al.

[Phys.Lett.B 842 \(2023\) 137563](#) | [arXiv:2206.07110](#) | [DOI:10.1016/j.physletb.2022.137563](#) | 2022-06-14

354. Observation of τ lepton pair production in ultraperipheral lead-lead collisions at $\sqrt{s_{\text{NN}}} = 5.02\text{ TeV}$

Tumasyan, Armen et al.

[Phys.Rev.Lett. 131 \(2023\) 151803](#) | [arXiv:2206.05192](#) | [DOI:10.1103/PhysRevLett.131.151803](#) | 2022-06-10

355. Search for Higgs boson decays into Z and J/ψ and for Higgs and Z boson decays into J/ψ or Y pairs in pp collisions at $\sqrt{s}=13\text{TeV}$

Tumasyan, Armen et al.

[Phys.Lett.B 842 \(2023\) 137534](#) | [arXiv:2206.03525](#) | [DOI:10.1016/j.physletb.2022.137534](#) | 2022-06-07

356. Observation of same-sign WW production from double parton scattering in proton-proton collisions at $\sqrt{s} = 13\text{ TeV}$

Tumasyan, Armen et al.

[Phys.Rev.Lett. 131 \(2023\) 091803](#) | [arXiv:2206.02681](#) | [DOI:10.1103/PhysRevLett.131.091803](#) | 2022-06-06

357. Combination of inclusive top-quark pair production cross-section measurements using ATLAS and CMS data at $\sqrt{s} = 7$ and 8 TeV

Aad, G. et al.

[JHEP 07 \(2023\) 213](#) | [arXiv:2205.13830](#) | [DOI:10.1007/JHEP07\(2023\)213](#) | 2022-05-27

358. Search for electroweak production of charginos and neutralinos at $\sqrt{s}=13\text{ TeV}$ in final states containing hadronic decays of WW, WZ, or WH and missing transverse momentum

Tumasyan, Armen et al.

[Phys.Lett.B 842 \(2023\) 137460](#) | [arXiv:2205.09597](#) | [DOI:10.1016/j.physletb.2022.137460](#) | 2022-05-19

359. Search for long-lived particles decaying to a pair of muons in proton-proton collisions at $\sqrt{s} = 13\text{ TeV}$

Tumasyan, Armen et al.

[JHEP 05 \(2023\) 228](#) | [arXiv:2205.08582](#) | [DOI:10.1007/JHEP05\(2023\)228](#) | 2022-05-17

360. Search for CP violating top quark couplings in pp collisions at $\sqrt{s} = 13\text{ TeV}$

Tumasyan, Armen et al.

[JHEP 07 \(2023\) 023](#) | [arXiv:2205.07434](#) | [DOI:10.1007/JHEP07\(2023\)023](#) | 2022-05-15

361. Search for heavy resonances and quantum black holes in $e\mu$, $e\tau$, and $\mu\tau$ final states in proton-proton collisions at $\sqrt{s} = 13\text{ TeV}$

Tumasyan, Armen et al.

[JHEP 05 \(2023\) 227](#) | [arXiv:2205.06709](#) | [DOI:10.1007/JHEP05\(2023\)227](#) | 2022-05-13

362. Search for Nonresonant Pair Production of Highly Energetic Higgs Bosons Decaying to Bottom Quarks

Tumasyan, Armen et al.

[Phys.Rev.Lett. 131 \(2023\) 041803](#) | [arXiv:2205.06667](#) | [DOI:10.1103/PhysRevLett.131.041803](#) | 2022-05-13

363. Search for Higgs Boson Decay to a Charm Quark-Antiquark Pair in Proton-Proton Collisions at $\sqrt{s}=13\text{TeV}$

Tumasyan, Armen et al.

[Phys.Rev.Lett. 131 \(2023\) 061801](#) | [arXiv:2205.05550](#) | [DOI:10.1103/PhysRevLett.131.061801](#) | 2022-05-11

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Tumasyan, Armen et al.

[Phys.Lett.B 841 \(2023\) 137495](#) | [arXiv:2205.05711](#) | [DOI:10.1016/j.physletb.2022.137495](#) | 2022-05-11

365. Measurement of the mass dependence of the transverse momentum of lepton pairs in Drell-Yan production in proton-proton collisions at $\sqrt{s} = 13$ TeV

Tumasyan, Armen et al.

[Eur.Phys.J.C 83 \(2023\) 628](#) | [arXiv:2205.04897](#) | [DOI:10.1140/epjc/s10052-023-11631-7](#) | 2022-05-10

366. Constraints on anomalous Higgs boson couplings to vector bosons and fermions from the production of Higgs bosons using the $\tau\tau$ final state

Tumasyan, Armen et al.

[Phys.Rev.D 108 \(2023\) 032013](#) | [arXiv:2205.05120](#) | [DOI:10.1103/PhysRevD.108.032013](#) | 2022-05-10

367. CMS pythia 8 colour reconnection tunes based on underlying-event data

Tumasyan, Armen et al.

[Eur.Phys.J.C 83 \(2023\) 587](#) | [arXiv:2205.02905](#) | [DOI:10.1140/epjc/s10052-023-11630-8](#) | 2022-05-05

368. Measurement of differential cross sections for the production of a Z boson in association with jets in proton-proton collisions at $\sqrt{s} = 13$ TeV

Tumasyan, Armen et al.

[Phys.Rev.D 108 \(2023\) 052004](#) | [arXiv:2205.02872](#) | [DOI:10.1103/PhysRevD.108.052004](#) | 2022-05-05

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Tumasyan, Armen et al.

[JHEP 06 \(2023\) 081](#) | [arXiv:2205.02314](#) | [DOI:10.1007/JHEP06\(2023\)081](#) | 2022-05-04

370. Search for narrow resonances in the b-tagged dijet mass spectrum in proton-proton collisions at $\sqrt{s}=13$ TeV

Tumasyan, Armen et al.

[Phys.Rev.D 108 \(2023\) 012009](#) | [arXiv:2205.01835](#) | [DOI:10.1103/PhysRevD.108.012009](#) | 2022-05-03

371. Strange hadron collectivity in pPb and PbPb collisions

Tumasyan, Armen et al.

[JHEP 05 \(2023\) 007](#) | [arXiv:2205.00080](#) | [DOI:10.1007/JHEP05\(2023\)007](#) | 2022-04-29

372. Azimuthal Correlations within Exclusive Dijets with Large Momentum Transfer in Photon-Lead Collisions

Tumasyan, Armen et al.

[Phys.Rev.Lett. 131 \(2023\) 051901](#) | [arXiv:2205.00045](#) | [DOI:10.1103/PhysRevLett.131.051901](#) | 2022-04-29

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[Phys.Lett.B 844 \(2023\) 137905](#) | [arXiv:2204.13486](#) | [DOI:10.1016/j.physletb.2023.137905](#) | 2022-04-28

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[Eur.Phys.J.C 83 \(2023\) 571](#) | [arXiv:2204.13532](#) | [DOI:10.1140/epjc/s10052-023-11581-0](#) | 2022-04-28

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[JHEP 05 \(2023\) 233](#) | [arXiv:2204.12945](#) | [DOI:10.1007/JHEP05\(2023\)233](#) | 2022-04-27

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Tumasyan, Armen et al.

[Eur.Phys.J.C 83 \(2023\) 562](#) | [arXiv:2204.12957](#) | [DOI:10.1140/epjc/s10052-023-11452-8](#) | 2022-04-27

377. Search for a massive scalar resonance decaying to a light scalar and a Higgs boson in the four b quarks final state with boosted topology

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[Phys.Lett.B 842 \(2023\) 137392](#) | [arXiv:2204.12413](#) | [DOI:10.1016/j.physletb.2022.137392](#) | 2022-04-26

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[Phys.Rev.D 108 \(2023\) 052002](#) | [arXiv:2204.12313](#) | [DOI:10.1103/PhysRevD.108.052002](#) | 2022-04-26

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Tumasyan, Armen et al.

[Phys.Lett.B 835 \(2022\) 137566](#) | [arXiv:2203.00480](#) | [DOI:10.1016/j.physletb.2022.137566](#) | 2022-03-01

380. Search for a W' boson decaying to a vector-like quark and a top or bottom quark in the all-jets final state at $\sqrt{s} = 13$ TeV

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[JHEP 09 \(2022\) 088](#) | [arXiv:2202.12988](#) | [DOI:10.1007/JHEP09\(2022\)088](#) | 2022-02-25

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[JHEP 2022 \(2022\) 063](#) | [arXiv:2202.12327](#) | [DOI:10.1007/JHEP08\(2022\)063](#) | 2022-02-24

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[Phys.Lett.B 835 \(2022\) 137397](#) | [arXiv:2202.11807](#) | [DOI:10.1016/j.physletb.2022.137397](#) | 2022-02-23

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Tumasyan, Armen et al.

[Phys.Rev.Lett. 129 \(2022\) 081802](#) | [arXiv:2202.09617](#) | [DOI:10.1103/PhysRevLett.129.081802](#) | 2022-02-19

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[Phys.Rev.D 105 \(2022\) 112007](#) | [arXiv:2202.08676](#) | [DOI:10.1103/PhysRevD.105.112007](#) | 2022-02-17

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[Nature Phys. 18 \(2022\) 1329](#) | [arXiv:2202.06923](#) | [DOI:10.1038/s41567-022-01682-0](#) | 2022-02-14

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Tumasyan, Armen et al.

[Phys.Rev.D 105 \(2022\) 092007](#) | [arXiv:2201.11585](#) | [DOI:10.1103/PhysRevD.105.092007](#) | 2022-01-27

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Tumasyan, Armen et al.

[Eur.Phys.J.C 82 \(2022\) 499](#) | [arXiv:2201.09131](#) | [DOI:10.1140/epjc/s10052-022-10315-y](#) | 2022-01-22

388. Search for resonances decaying to three W bosons in proton-proton collisions at $\sqrt{s} = 13$ TeV

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[Phys.Rev.Lett. 129 \(2022\) 021802](#) | [arXiv:2201.08476](#) | [DOI:10.1103/PhysRevLett.129.021802](#) | 2022-01-20

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[JINST 17 \(2022\) P07023](#) | [arXiv:2201.08458](#) | [DOI:10.1088/1748-0221/17/07/P07023](#) | 2022-01-20

390. Precision measurement of the W boson decay branching fractions in proton-proton collisions at $\sqrt{s} = 13$ TeV

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[Phys.Rev.D 105 \(2022\) 072008](#) | [arXiv:2201.07861](#) | [DOI:10.1103/PhysRevD.105.072008](#) | 2022-01-19

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[JHEP 06 \(2022\) 082](#) | [arXiv:2201.07859](#) | [DOI:10.1007/JHEP06\(2022\)082](#) | 2022-01-19

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[JHEP 05 \(2022\) 091](#) | [arXiv:2201.07301](#) | [DOI:10.1007/JHEP05\(2022\)091](#) | 2022-01-18

393. Search for long-lived heavy neutral leptons with displaced vertices in proton-proton collisions at $\sqrt{s} = 13$ TeV

Tumasyan, Armen et al.

[JHEP 07 \(2022\) 081](#) | [arXiv:2201.05578](#) | [DOI:10.1007/JHEP07\(2022\)081](#) | 2022-01-14

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[JHEP 05 \(2022\) 014](#) | [arXiv:2201.04206](#) | [DOI:10.1007/JHEP05\(2022\)014](#) | 2022-01-11

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Tumasyan, Armen et al.

[Phys.Rev.Lett. 128 \(2022\) 252301](#) | [arXiv:2201.02659](#) | [DOI:10.1103/PhysRevLett.128.252301](#) | 2022-01-07

396. Search for high-mass resonances decaying to a jet and a Lorentz-boosted resonance in proton-proton collisions at $\sqrt{s} = 13$ TeV

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[Phys.Lett.B 832 \(2022\) 137263](#) | [arXiv:2201.02140](#) | [DOI:10.1016/j.physletb.2022.137263](#) | 2022-01-06

397. Search for single production of a vector-like T quark decaying to a top quark and a Z boson in the final state with jets and missing transverse momentum at $\sqrt{s} = 13$ TeV

Tumasyan, Armen et al.

[JHEP 05 \(2022\) 093](#) | [arXiv:2201.02227](#) | [DOI:10.1007/JHEP05\(2022\)093](#) | 2022-01-06

398. Search for long-lived particles decaying into muon pairs in proton-proton collisions at $\sqrt{s} = 13$ TeV collected with a dedicated high-rate data stream

Tumasyan, Armen et al.

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399. Search for resonances decaying to three W bosons in the hadronic final state in proton-proton collisions at $\sqrt{s} = 13$ TeV

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[Phys.Rev.D 106 \(2022\) 012002](#) | [arXiv:2112.13090](#) | [DOI:10.1103/PhysRevD.106.012002](#) | 2021-12-24

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[Phys.Rev.Lett. 129 \(2022\) 022001](#) | [arXiv:2112.12236](#) | [DOI:10.1103/PhysRevLett.129.022001](#) | 2021-12-22

401. Search for resonant production of strongly coupled dark matter in proton-proton collisions at 13 TeV

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[JHEP 06 \(2022\) 156](#) | [arXiv:2112.11125](#) | [DOI:10.1007/JHEP06\(2022\)156](#) | 2021-12-21

402. Search for flavor-changing neutral current interactions of the top quark and the Higgs boson decaying to a bottom quark-antiquark pair at $\sqrt{s} = 13$ TeV

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[JHEP 02 \(2022\) 169](#) | [arXiv:2112.09734](#) | [DOI:10.1007/JHEP02\(2022\)169](#) | 2021-12-17

403. Measurement of the production cross section for Z+b jets in proton-proton collisions at $\sqrt{s} = 13$ TeV

Tumasyan, Armen et al.

[Phys.Rev.D 105 \(2022\) 092014](#) | [arXiv:2112.09659](#) | [DOI:10.1103/PhysRevD.105.092014](#) | 2021-12-17

404. Measurement of the inclusive tt production cross section in proton-proton collisions at $\sqrt{s} = 5.02$ TeV

Tumasyan, Armen et al.

[JHEP 04 \(2022\) 144](#) | [arXiv:2112.09114](#) | [DOI:10.1007/JHEP04\(2022\)144](#) | 2021-12-16

405. Evidence for WW/WZ vector boson scattering in the decay channel $\ell\nu qq$ produced in association with two jets in proton-proton collisions at $\sqrt{s}=13$ TeV

Tumasyan, Armen et al.

[Phys.Lett.B 834 \(2022\) 137438](#) | [arXiv:2112.05259](#) | [DOI:10.1016/j.physletb.2022.137438](#) | 2021-12-09

406. Search for a right-handed W boson and a heavy neutrino in proton-proton collisions at $\sqrt{s} = 13$ TeV

Tumasyan, Armen et al.

[JHEP 04 \(2022\) 047](#) | [arXiv:2112.03949](#) | [DOI:10.1007/JHEP04\(2022\)047](#) | 2021-12-07

407. Search for heavy resonances decaying to a pair of Lorentz-boosted Higgs bosons in final states with leptons and a bottom quark pair at $\sqrt{s}= 13$ TeV

Tumasyan, Armen et al.

[JHEP 05 \(2022\) 005](#) | [arXiv:2112.03161](#) | [DOI:10.1007/JHEP05\(2022\)005](#) | 2021-12-06

408. Measurements of the associated production of a W boson and a charm quark in proton-proton collisions at $\sqrt{s}=8$ TeV

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[Eur.Phys.J.C 82 \(2022\) 1094](#) | [arXiv:2112.00895](#) | [DOI:10.1140/epjc/s10052-022-10897-7](#) | 2021-12-01

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[Phys.Rev.D 105 \(2022\) 052003](#) | [arXiv:2111.13948](#) | [DOI:10.1103/PhysRevD.105.052003](#) | 2021-11-27

410. Search for heavy resonances decaying to ZZ or ZW and axion-like particles mediating nonresonant ZZ or ZH production at $\sqrt{s} = 13$ TeV

Tumasyan, Armen et al.

[JHEP 04 \(2022\) 087](#) | [arXiv:2111.13669](#) | [DOI:10.1007/JHEP04\(2022\)087](#) | 2021-11-26

411. Search for a heavy resonance decaying into a top quark and a W boson in the lepton+jets final state at $\sqrt{s} = 13$ TeV

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[JHEP 04 \(2022\) 048](#) | [arXiv:2111.10216](#) | [DOI:10.1007/JHEP04\(2022\)048](#) | 2021-11-19

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[JHEP 02 \(2022\) 142](#) | [arXiv:2111.10431](#) | [DOI:10.1007/JHEP02\(2022\)142](#) | 2021-11-19

413. Strategies and performance of the CMS silicon tracker alignment during LHC Run 2

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[Nucl.Instrum.Meth.A 1037 \(2022\) 166795](#) | [arXiv:2111.08757](#) | [DOI:10.1016/j.nima.2022.166795](#) | 2021-11-16

414. Search for supersymmetry in final states with two or three soft leptons and missing transverse momentum in proton-proton collisions at $\sqrt{s} = 13$ TeV

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[JHEP 04 \(2022\) 091](#) | [arXiv:2111.06296](#) | [DOI:10.1007/JHEP04\(2022\)091](#) | 2021-11-11

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[Nature Phys. 19 \(2023\) 338](#) | [arXiv:2111.05370](#) | [DOI:10.1038/s41567-022-01838-y](#) | 2021-11-09

416. Study of dijet events with large rapidity separation in proton-proton collisions at $\sqrt{s} = 2.76$ TeV

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[JHEP 03 \(2022\) 189](#) | [arXiv:2111.04605](#) | [DOI:10.1007/JHEP03\(2022\)189](#) | 2021-11-08

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[JINST 17 \(2022\) P03014](#) | [arXiv:2111.03027](#) | [DOI:10.1088/1748-0221/17/03/P03014](#) | 2021-11-04

418. Inclusive and differential cross section measurements of single top quark production in association with a Z boson in proton-proton collisions at $\sqrt{s} = 13$ TeV

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[JHEP 02 \(2022\) 107](#) | [arXiv:2111.02860](#) | [DOI:10.1007/JHEP02\(2022\)107](#) | 2021-11-04

419. Search for Flavor-Changing Neutral Current Interactions of the Top Quark and Higgs Boson in Final States with Two Photons in Proton-Proton Collisions at $\sqrt{s} = 13$ TeV

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[Phys.Rev.Lett. 129 \(2022\) 032001](#) | [arXiv:2111.02219](#) | [DOI:10.1103/PhysRevLett.129.032001](#) | 2021-11-03

420. Search for low-mass dilepton resonances in Higgs boson decays to four-lepton final states in proton-proton collisions at $\sqrt{s} = 13$ TeV

Tumasyan, A. et al.

[Eur.Phys.J.C 82 \(2022\) 290](#) | [arXiv:2111.01299](#) | [DOI:10.1140/epjc/s10052-022-10127-0](#) | 2021-11-01

421. Search for long-lived particles produced in association with a Z boson in proton-proton collisions at $\sqrt{s} = 13$ TeV

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[JHEP 03 \(2022\) 160](#) | [arXiv:2110.13218](#) | [DOI:10.1007/JHEP03\(2022\)160](#) | 2021-10-25

422. Measurement of the inclusive and differential WZ production cross sections, polarization angles, and triple gauge couplings in pp collisions at $\sqrt{s} = 13$ TeV

Tumasyan, Armen et al.

[JHEP 07 \(2022\) 032](#) | [arXiv:2110.11231](#) | [DOI:10.1007/JHEP07\(2022\)032](#) | 2021-10-21

423. First Search for Exclusive Diphoton Production at High Mass with Tagged Protons in Proton-Proton Collisions at $\sqrt{s} = 13$ TeV

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[Phys.Rev.Lett. 129 \(2022\) 011801](#) | [arXiv:2110.05916](#) | [DOI:10.1103/PhysRevLett.129.011801](#) | 2021-10-12

424. Search for long-lived particles decaying to leptons with large impact parameter in proton-proton collisions at $\sqrt{s} = 13$ TeV

Tumasyan, Armen et al.

[Eur.Phys.J.C 82 \(2022\) 153](#) | [arXiv:2110.04809](#) | [DOI:10.1140/epjc/s10052-022-10027-3](#) | 2021-10-10

425. Analysis of the CP structure of the Yukawa coupling between the Higgs boson and τ leptons in proton-proton collisions at $\sqrt{s} = 13$ TeV

Tumasyan, Armen et al.

[JHEP 06 \(2022\) 012](#) | [arXiv:2110.04836](#) | [DOI:10.1007/JHEP06\(2022\)012](#) | 2021-10-10

426. Measurement of double-parton scattering in inclusive production of four jets with low transverse momentum in proton-proton collisions at $\sqrt{s} = 13$ TeV

Tumasyan, Armen et al.

[JHEP 01 \(2022\) 177](#) | [arXiv:2109.13822](#) | [DOI:10.1007/JHEP01\(2022\)177](#) | 2021-09-28

427. Search for heavy resonances decaying to $Z(\nu\bar{\nu})V(q\bar{q}')$ in proton-proton collisions at $\sqrt{s} = 13$ TeV

Tumasyan, Armen et al.

[Phys.Rev.D 106 \(2022\) 012004](#) | [arXiv:2109.08268](#) | [DOI:10.1103/PhysRevD.106.012004](#) | 2021-09-17

428. Search for heavy resonances decaying to WW, WZ, or WH boson pairs in the lepton plus merged jet final state in proton-proton collisions at $\sqrt{s} = 13$ TeV

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[Phys.Rev.D 105 \(2022\) 032008](#) | [arXiv:2109.06055](#) | [DOI:10.1103/PhysRevD.105.032008](#) | 2021-09-13

429. Study of quark and gluon jet substructure in Z+jet and dijet events from pp collisions

Tumasyan, Armen et al.

[JHEP 01 \(2022\) 188](#) | [arXiv:2109.03340](#) | [DOI:10.1007/JHEP01\(2022\)188](#) | 2021-09-07

430. Observation of B^0_s mesons and measurement of the B^0_s/B^+ yield ratio in PbPb collisions at $\sqrt{s_{NN}} = 5.02$ TeV

Tumasyan, Armen et al.

[Phys.Lett.B 829 \(2022\) 137062](#) | [arXiv:2109.01908](#) | [DOI:10.1016/j.physletb.2022.137062](#) | 2021-09-04

431. Observation of tW production in the single-lepton channel in pp collisions at $\sqrt{s} = 13$ TeV

Tumasyan, Armen et al.

[JHEP 11 \(2021\) 111](#) | [arXiv:2109.01706](#) | [DOI:10.1007/JHEP11\(2021\)111](#) | 2021-09-03

432. Measurement of the top quark mass using events with a single reconstructed top quark in pp collisions at $\sqrt{s} = 13$ TeV

Tumasyan, Armen et al.

[JHEP 12 \(2021\) 161](#) | [arXiv:2108.10407](#) | [DOI:10.1007/JHEP12\(2021\)161](#) | 2021-08-23

433. Measurement of differential $t\bar{t}$ production cross sections in the full kinematic range using lepton+jets events from proton-proton collisions at $\sqrt{s} = 13$ TeV

Tumasyan, Armen et al.

[Phys.Rev.D 104 \(2021\) 092013](#) | [arXiv:2108.02803](#) | [DOI:10.1103/PhysRevD.104.092013](#) | 2021-08-05

434. Probing effective field theory operators in the associated production of top quarks with a Z boson in multilepton final states at $\sqrt{s} = 13$ TeV

Lee, Kyeongpil et al.

[JHEP 12 \(2021\) 083](#) | [arXiv:2107.13896](#) | [DOI:10.1007/JHEP12\(2021\)083](#) | 2021-07-29

435. Search for new particles in events with energetic jets and large missing transverse momentum in proton-proton collisions at $\sqrt{s} = 13$ TeV

Tumasyan, Armen et al.

[JHEP 11 \(2021\) 153](#) | [arXiv:2107.13021](#) | [DOI:10.1007/JHEP11\(2021\)153](#) | 2021-07-27

436. Search for chargino-neutralino production in events with Higgs and W bosons using 137 fb^{-1} of proton-proton collisions at $\sqrt{s} = 13$ TeV

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[JHEP 10 \(2021\) 045](#) | [arXiv:2107.12553](#) | [DOI:10.1007/JHEP10\(2021\)045](#) | 2021-07-27

437. Measurement of the inclusive and differential Higgs boson production cross sections in the decay mode to a pair of τ leptons in pp collisions at $\sqrt{s} = 13$ TeV

Tumasyan, Armen et al.

[Phys.Rev.Lett. 128 \(2022\) 081805](#) | [arXiv:2107.11486](#) | [DOI:10.1103/PhysRevLett.128.081805](#) | 2021-07-23

438. Combined searches for the production of supersymmetric top quark partners in proton-proton collisions at $\sqrt{s} = 13$ TeV

Tumasyan, Armen et al.

[Eur.Phys.J.C 81 \(2021\) 970](#) | [arXiv:2107.10892](#) | [DOI:10.1140/epjc/s10052-021-09721-5](#) | 2021-07-22

439. Search for Long-Lived Particles Decaying in the CMS End Cap Muon Detectors in Proton-Proton Collisions at $\sqrt{s} = 13$ TeV

Tumasyan, Armen et al.

[Phys.Rev.Lett. 127 \(2021\) 261804](#) | [arXiv:2107.04838](#) | [DOI:10.1103/PhysRevLett.127.261804](#) | 2021-07-10

440. Measurement of the inclusive and differential tt γ cross sections in the single-lepton channel and EFT interpretation at $\sqrt{s} = 13$ TeV

Tumasyan, Armen et al.

[JHEP 12 \(2021\) 180](#) | [arXiv:2107.01508](#) | [DOI:10.1007/JHEP12\(2021\)180](#) | 2021-07-03

441. Measurement of prompt open-charm production cross sections in proton-proton collisions at $\sqrt{s} = 13$ TeV

Tumasyan, Armen et al.

[JHEP 11 \(2021\) 225](#) | [arXiv:2107.01476](#) | [DOI:10.1007/JHEP11\(2021\)225](#) | 2021-07-03

442. Measurements of the electroweak diboson production cross sections in proton-proton collisions at $\sqrt{s} = 5.02$ TeV using leptonic decays

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[Phys.Rev.Lett. 127 \(2021\) 191801](#) | [arXiv:2107.01137](#) | [DOI:10.1103/PhysRevLett.127.191801](#) | 2021-07-02

443. Search for electroweak production of charginos and neutralinos in proton-proton collisions at $\sqrt{s} = 13$ TeV

Tumasyan, Armen et al.

[JHEP 04 \(2022\) 147](#) | [arXiv:2106.14246](#) | [DOI:10.1007/JHEP04\(2022\)147](#) | 2021-06-27

444. Fragmentation of jets containing a prompt J/ ψ meson in PbPb and pp collisions at $\sqrt{s_{NN}} = 5.02$ TeV

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[Phys.Lett.B 825 \(2022\) 136842](#) | [arXiv:2106.13235](#) | [DOI:10.1016/j.physletb.2021.136842](#) | 2021-06-23

445. Measurement of the electroweak production of $Z\gamma$ and two jets in proton-proton collisions at $\sqrt{s} = 13$ TeV and constraints on anomalous quartic gauge couplings

Tumasyan, Armen et al.

[Phys.Rev.D 104 \(2021\) 072001](#) | [arXiv:2106.11082](#) | [DOI:10.1103/PhysRevD.104.072001](#) | 2021-06-21

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[Phys.Lett.B 826 \(2022\) 136888](#) | [arXiv:2106.10509](#) | [DOI:10.1016/j.physletb.2022.136888](#) | 2021-06-19

447. Search for a heavy Higgs boson decaying into two lighter Higgs bosons in the $\tau\tau b\bar{b}$ final state at 13 TeV

Tumasyan, Armen et al.

[JHEP 11 \(2021\) 057](#) | [arXiv:2106.10361](#) | [DOI:10.1007/JHEP11\(2021\)057](#) | 2021-06-18

448. Study of Z boson plus jets events using variables sensitive to double-parton scattering in pp collisions at 13 TeV

Tumasyan, Armen et al.

[JHEP 10 \(2021\) 176](#) | [arXiv:2105.14511](#) | [DOI:10.1007/JHEP10\(2021\)176](#) | 2021-05-30

449. Measurements of the $pp \rightarrow W^\pm \gamma \gamma$ and $pp \rightarrow Z \gamma \gamma$ cross sections at $\sqrt{s} = 13$ TeV and limits on anomalous quartic gauge couplings

Tumasyan, Armen et al.

[JHEP 10 \(2021\) 174](#) | [arXiv:2105.12780](#) | [DOI:10.1007/JHEP10\(2021\)174](#) | 2021-05-26

450. Search for strongly interacting massive particles generating trackless jets in proton-proton collisions at $\sqrt{s} = 13$ TeV

Tumasyan, Armen et al.

[Eur.Phys.J.C 82 \(2022\) 213](#) | [arXiv:2105.09178](#) | [DOI:10.1140/epjc/s10052-022-10095-5](#) | 2021-05-19

451. Search for lepton-flavor violating decays of the Higgs boson in the $\mu\tau$ and $e\tau$ final states in proton-proton collisions at $\sqrt{s} = 13$ TeV

Sirunyan, Albert M et al.

[Phys.Rev.D 104 \(2021\) 032013](#) | [arXiv:2105.03007](#) | [DOI:10.1103/PhysRevD.104.032013](#) | 2021-05-06

452. Search for long-lived particles decaying to jets with displaced vertices in proton-proton collisions at $\sqrt{s} = 13$ TeV

Sirunyan, A.M. et al.

[Phys.Rev.D 104 \(2021\) 052011](#) | [arXiv:2104.13474](#) | [DOI:10.1103/PhysRevD.104.052011](#) | 2021-04-27

453. Search for a heavy resonance decaying to a top quark and a W boson at $\sqrt{s} = 13$ TeV in the fully hadronic final state

Sirunyan, Albert M et al.

[JHEP 12 \(2021\) 106](#) | [arXiv:2104.12853](#) | [DOI:10.1007/JHEP12\(2021\)106](#) | 2021-04-26

454. Constraints on anomalous Higgs boson couplings to vector bosons and fermions in its production and decay using the four-lepton final state

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[Phys.Rev.D 104 \(2021\) 052004](#) | [arXiv:2104.12152](#) | [DOI:10.1103/PhysRevD.104.052004](#) | 2021-04-25

455. Search for W' bosons decaying to a top and a bottom quark at $\sqrt{s}=13$ TeV in the hadronic final state

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[Phys.Lett.B 820 \(2021\) 136535](#) | [arXiv:2104.04831](#) | [DOI:10.1016/j.physletb.2021.136535](#) | 2021-04-10

456. Search for charged Higgs bosons produced in vector boson fusion processes and decaying into vector boson pairs in proton-proton collisions at $\sqrt{s} = 13$ TeV

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[Eur.Phys.J.C 81 \(2021\) 723](#) | [arXiv:2104.04762](#) | [DOI:10.1140/epjc/s10052-021-09472-3](#) | 2021-04-10

457. Precision luminosity measurement in proton-proton collisions at $\sqrt{s} = 13$ TeV in 2015 and 2016 at CMS

Sirunyan, Albert M et al.

[Eur.Phys.J.C 81 \(2021\) 800](#) | [arXiv:2104.01927](#) | [DOI:10.1140/epjc/s10052-021-09538-2](#) | 2021-04-05

458. Constraints on the Initial State of Pb-Pb Collisions via Measurements of Z-Boson Yields and Azimuthal Anisotropy at $\sqrt{s_{NN}}=5.02$ TeV

Sirunyan, Albert M et al.

[Phys.Rev.Lett. 127 \(2021\) 102002](#) | [arXiv:2103.14089](#) | [DOI:10.1103/PhysRevLett.127.102002](#) | 2021-03-25

459. Measurements of Higgs boson production cross sections and couplings in the diphoton decay channel at $\sqrt{s} = 13$ TeV

Sirunyan, Albert M et al.

[JHEP 07 \(2021\) 027](#) | [arXiv:2103.06956](#) | [DOI:10.1007/JHEP07\(2021\)027](#) | 2021-03-11

460. Measurements of production cross sections of the Higgs boson in the four-lepton final state in proton-proton collisions at $\sqrt{s} = 13$ TeV

Sirunyan, Albert M et al.

[Eur.Phys.J.C 81 \(2021\) 488](#) | [arXiv:2103.04956](#) | [DOI:10.1140/epjc/s10052-021-09200-x](#) | 2021-03-08

461. Using Z Boson Events to Study Parton-Medium Interactions in Pb-Pb Collisions

Sirunyan, Albert M et al.

[Phys.Rev.Lett. 128 \(2022\) 122301](#) | [arXiv:2103.04377](#) | [DOI:10.1103/PhysRevLett.128.122301](#) | 2021-03-07

462. Search for resonant and nonresonant new phenomena in high-mass dilepton final states at $\sqrt{s} = 13$ TeV

Sirunyan, Albert M et al.

[JHEP 07 \(2021\) 208](#) | [arXiv:2103.02708](#) | [DOI:10.1007/JHEP07\(2021\)208](#) | 2021-03-03

463. Search for top squark production in fully-hadronic final states in proton-proton collisions at $\sqrt{s} = 13$ TeV

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[Phys.Rev.D 104 \(2021\) 052001](#) | [arXiv:2103.01290](#) | [DOI:10.1103/PhysRevD.104.052001](#) | 2021-03-01

464. Study of Drell-Yan dimuon production in proton-lead collisions at $\sqrt{s_{NN}} = 8.16$ TeV

Sirunyan, Albert M et al.

[JHEP 05 \(2021\) 182](#) | [arXiv:2102.13648](#) | [DOI:10.1007/JHEP05\(2021\)182](#) | 2021-02-26

465. First measurement of large area jet transverse momentum spectra in heavy-ion collisions

Sirunyan, Albert M et al.

[JHEP 05 \(2021\) 284](#) | [arXiv:2102.13080](#) | [DOI:10.1007/JHEP05\(2021\)284](#) | 2021-02-25

466. Evidence for X(3872) in Pb-Pb Collisions and Studies of its Prompt Production at $\sqrt{s_{NN}}=5.02$ TeV

Sirunyan, Albert M. et al.

[Phys.Rev.Lett. 128 \(2022\) 032001](#) | [arXiv:2102.13048](#) | [DOI:10.1103/PhysRevLett.128.032001](#) | 2021-02-25

467. Measurements of angular distance and momentum ratio distributions in three-jet and Z + two-jet final states in pp collisions

Sirunyan, Albert M et al.

[Eur.Phys.J.C 81 \(2021\) 852](#) | [arXiv:2102.08816](#) | [DOI:10.1140/epjc/s10052-021-09570-2](#) | 2021-02-17

468. Search for a heavy vector resonance decaying to a Z^0 boson and a Higgs boson in proton-proton collisions at $\sqrt{s} = 13$ TeV

Sirunyan, Albert M et al.

[Eur.Phys.J.C 81 \(2021\) 688](#) | [arXiv:2102.08198](#) | [DOI:10.1140/epjc/s10052-021-09348-6](#) | 2021-02-16

469. Search for top squarks in final states with two top quarks and several light-flavor jets in proton-proton collisions at $\sqrt{s} = 13$ TeV

Sirunyan, Albert M et al.

[Phys.Rev.D 104 \(2021\) 032006](#) | [arXiv:2102.06976](#) | [DOI:10.1103/PhysRevD.104.032006](#) | 2021-02-13

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Sirunyan, Albert M et al.

[Phys.Rev.D 104 \(2021\) 032009](#) | [arXiv:2102.06945](#) | [DOI:10.1103/PhysRevD.104.032009](#) | 2021-02-13

471. Performance of the CMS muon trigger system in proton-proton collisions at $\sqrt{s} = 13$ TeV

Sirunyan, Albert M et al.

[JINST 16 \(2021\) P07001](#) | [arXiv:2102.04790](#) | [DOI:10.1088/1748-0221/16/07/P07001](#) | 2021-02-09

472. Observation of a New Excited Beauty Strange Baryon Decaying to $\Xi^- \pi^+ \pi^-$

Sirunyan, Albert M et al.

[Phys.Rev.Lett. 126 \(2021\) 252003](#) | [arXiv:2102.04524](#) | [DOI:10.1103/PhysRevLett.126.252003](#) | 2021-02-08

473. Measurement of the $W\gamma$ Production Cross Section in Proton-Proton Collisions at $\sqrt{s}=13$ TeV and Constraints on Effective Field Theory Coefficients

Sirunyan, Albert M et al.

[Phys.Rev.Lett. 126 \(2021\) 252002](#) | [arXiv:2102.02283](#) | [DOI:10.1103/PhysRevLett.126.252002](#) | 2021-02-03

474. Measurements of the differential cross sections of the production of $Z + \text{jets}$ and $\gamma + \text{jets}$ and of Z boson emission collinear with a jet in pp collisions at $\sqrt{s} = 13$ TeV

Sirunyan, Albert M et al.

[JHEP 05 \(2021\) 285](#) | [arXiv:2102.02238](#) | [DOI:10.1007/JHEP05\(2021\)285](#) | 2021-02-03

475. In-medium modification of dijets in PbPb collisions at $\sqrt{s_{NN}} = 5.02$ TeV

Sirunyan, Albert M et al.

[JHEP 05 \(2021\) 116](#) | [arXiv:2101.04720](#) | [DOI:10.1007/JHEP05\(2021\)116](#) | 2021-01-12

476. Measurement of the Z boson differential production cross section using its invisible decay mode ($Z\nu\bar{\nu}$) in proton-proton collisions at $\sqrt{s}=13$ TeV

Sirunyan, Albert M et al.

[JHEP 05 \(2021\) 205](#) | [arXiv:2012.09254](#) | [DOI:10.1007/JHEP05\(2021\)205](#) | 2020-12-16

477. First measurement of the cross section for top quark pair production with additional charm jets using dileptonic final states in pp collisions at $s=13\text{TeV}$

Sirunyan, Albert M et al.

[Phys.Lett.B 820 \(2021\) 136565](#) | [arXiv:2012.09225](#) | [DOI:10.1016/j.physletb.2021.136565](#) | 2020-12-16

478. Search for supersymmetry in final states with two oppositely charged same-flavor leptons and missing transverse momentum in proton-proton collisions at $\sqrt{s} = 13$ TeV

Sirunyan, Albert M et al.

[JHEP 04 \(2021\) 123](#) | [arXiv:2012.08600](#) | [DOI:10.1007/JHEP04\(2021\)123](#) | 2020-12-15

479. Electron and photon reconstruction and identification with the CMS experiment at the CERN LHC

Sirunyan, Albert M et al.

[JINST 16 \(2021\) P05014](#) | [arXiv:2012.06888](#) | [DOI:10.1088/1748-0221/16/05/P05014](#) | 2020-12-12

480. Measurement of differential cross sections for Z bosons produced in association with charm jets in pp collisions at $\sqrt{s} = 13$ TeV

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[JHEP 04 \(2021\) 109](#) | [arXiv:2012.04119](#) | [DOI:10.1007/JHEP04\(2021\)109](#) | 2020-12-07

481. Search for new physics in top quark production with additional leptons in proton-proton collisions at $\sqrt{s} = 13$ TeV using effective field theory

Sirunyan, Albert M et al.

[JHEP 03 \(2021\) 095](#) | [arXiv:2012.04120](#) | [DOI:10.1007/JHEP03\(2021\)095](#) | 2020-12-07

482. Search for singly and pair-produced leptoquarks coupling to third-generation fermions in proton-proton collisions at $\sqrt{s}=13\text{TeV}$

Sirunyan, Albert M et al.

[Phys.Lett.B 819 \(2021\) 136446](#) | [arXiv:2012.04178](#) | [DOI:10.1016/j.physletb.2021.136446](#) | 2020-12-07

483. Search for long-lived particles using displaced jets in proton-proton collisions at $\sqrt{s} = 13\text{ TeV}$

Sirunyan, Albert M et al.

[Phys.Rev.D 104 \(2021\) 012015](#) | [arXiv:2012.01581](#) | [DOI:10.1103/PhysRevD.104.012015](#) | 2020-12-02

484. Search for nonresonant Higgs boson pair production in final states with two bottom quarks and two photons in proton-proton collisions at $\sqrt{s} = 13\text{ TeV}$

Sirunyan, Albert M et al.

[JHEP 03 \(2021\) 257](#) | [arXiv:2011.12373](#) | [DOI:10.1007/JHEP03\(2021\)257](#) | 2020-11-24

485. Search for the rare decay of the W boson into a pion and a photon in proton-proton collisions at $\sqrt{s}=13\text{ TeV}$

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[Phys.Lett.B 819 \(2021\) 136409](#) | [arXiv:2011.06028](#) | [DOI:10.1016/j.physletb.2021.136409](#) | 2020-11-11

486. Observation of Forward Neutron Multiplicity Dependence of Dimuon Acoplanarity in Ultraperipheral Pb-Pb Collisions at $\sqrt{s_{\text{NN}}}=5.02\text{ TeV}$

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[Phys.Rev.Lett. 127 \(2021\) 122001](#) | [arXiv:2011.05239](#) | [DOI:10.1103/PhysRevLett.127.122001](#) | 2020-11-10

487. Development and validation of HERWIG 7 tunes from CMS underlying-event measurements

Sirunyan, Albert M et al.

[Eur.Phys.J.C 81 \(2021\) 312](#) | [arXiv:2011.03422](#) | [DOI:10.1140/epjc/s10052-021-08949-5](#) | 2020-11-06

488. Measurement of the Higgs boson production rate in association with top quarks in final states with electrons, muons, and hadronically decaying tau leptons at $\sqrt{s} = 13\text{ TeV}$

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[Eur.Phys.J.C 81 \(2021\) 378](#) | [arXiv:2011.03652](#) | [DOI:10.1140/epjc/s10052-021-09014-x](#) | 2020-11-06

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Khachatryan, Vardan et al.

[JINST 16 \(2021\) P02010](#) | [arXiv:2011.01185](#) | [DOI:10.1088/1748-0221/16/02/P02010](#) | 2020-11-02

490. Angular analysis of the decay $B^+ \rightarrow K^{*(892)^+} \mu^+ \mu^-$ in proton-proton collisions at $\sqrt{s} = 8\text{ TeV}$

Sirunyan, Albert M et al.

[JHEP 04 \(2021\) 124](#) | [arXiv:2010.13968](#) | [DOI:10.1007/JHEP04\(2021\)124](#) | 2020-10-26

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Sirunyan, Albert M et al.

[Eur.Phys.J.C 81 \(2021\) 629](#) | [arXiv:2010.02984](#) | [DOI:10.1140/epjc/s10052-021-09236-z](#) | 2020-10-06

492. Search for dark photons in Higgs boson production via vector boson fusion in proton-proton collisions at $\sqrt{s} = 13\text{ TeV}$

Sirunyan, Albert M et al.

[JHEP 03 \(2021\) 011](#) | [arXiv:2009.14009](#) | [DOI:10.1007/JHEP03\(2021\)011](#) | 2020-09-29

493. Measurement of prompt D^0 and $\bar{D}^0$ meson azimuthal anisotropy and search for strong electric fields in PbPb collisions at $\sqrt{s_{\text{NN}}} = 5.02\text{ TeV}$

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[Phys.Lett.B 816 \(2021\) 136253](#) | [arXiv:2009.12628](#) | [DOI:10.1016/j.physletb.2021.136253](#) | 2020-09-26

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[Phys.Lett.B 812 \(2021\) 136018](#) | [arXiv:2009.09429](#) | [DOI:10.1016/j.physletb.2020.136018](#) | 2020-09-20

495. Measurement of the top quark Yukawa coupling from $t\bar{t}$ kinematic distributions in the dilepton final state in proton-proton collisions at $\sqrt{s} = 13$ TeV

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[Phys.Rev.D 102 \(2020\) 092013](#) | [arXiv:2009.07123](#) | [DOI:10.1103/PhysRevD.102.092013](#) | 2020-09-15

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Sirunyan, Albert M et al.

[Phys.Lett.B 813 \(2021\) 136036](#) | [arXiv:2009.07065](#) | [DOI:10.1016/j.physletb.2020.136036](#) | 2020-09-15

497. Evidence for Higgs boson decay to a pair of muons

Sirunyan, Albert M et al.

[JHEP 01 \(2021\) 148](#) | [arXiv:2009.04363](#) | [DOI:10.1007/JHEP01\(2021\)148](#) | 2020-09-09

498. Measurements of $pp \rightarrow ZZ$ production cross sections and constraints on anomalous triple gauge couplings at $\sqrt{s} = 13$ TeV

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[Eur.Phys.J.C 81 \(2021\) 200](#) | [arXiv:2009.01186](#) | [DOI:10.1140/epjc/s10052-020-08817-8](#) | 2020-09-02

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[Phys.Rev.D 102 \(2020\) 092001](#) | [arXiv:2009.00119](#) | [DOI:10.1103/PhysRevD.102.092001](#) | 2020-08-31

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[Phys.Lett.B 811 \(2020\) 135988](#) | [arXiv:2008.10521](#) | [DOI:10.1016/j.physletb.2020.135988](#) | 2020-08-24

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[Phys.Rev.D 102 \(2020\) 112004](#) | [arXiv:2008.09835](#) | [DOI:10.1103/PhysRevD.102.112004](#) | 2020-08-22

502. Measurement of $B_c(2S)^+$ and $B_c^{*}(2S)^+$ cross section ratios in proton-proton collisions at $\sqrt{s} = 13$ TeV

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[Phys.Rev.D 102 \(2020\) 092007](#) | [arXiv:2008.08629](#) | [DOI:10.1103/PhysRevD.102.092007](#) | 2020-08-19

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[Phys.Rev.D 103 \(2021\) 052008](#) | [arXiv:2008.07860](#) | [DOI:10.1103/PhysRevD.103.052008](#) | 2020-08-18

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[Phys.Lett.B 812 \(2021\) 135992](#) | [arXiv:2008.07013](#) | [DOI:10.1016/j.physletb.2020.135992](#) | 2020-08-16

505. Search for top squark pair production using dilepton final states in pp collision data collected at $\sqrt{s} = 13$ TeV

Sirunyan, Albert M et al.

[Eur.Phys.J.C 81 \(2021\) 3](#) | [arXiv:2008.05936](#) | [DOI:10.1140/epjc/s10052-020-08701-5](#) | 2020-08-13

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[Eur.Phys.J.C 81 \(2021\) 13](#) | [arXiv:2008.04735](#) | [DOI:10.1140/epjc/s10052-020-08739-5](#) | 2020-08-11

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[Phys.Rev.D 102 \(2020\) 092012](#) | [arXiv:2008.04174](#) | [DOI:10.1103/PhysRevD.102.092012](#) | 2020-08-10

508. Search for supersymmetry in proton-proton collisions at $\sqrt{s} = 13$ TeV in events with high-momentum Z bosons and missing transverse momentum

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[JHEP 09 \(2020\) 149](#) | [arXiv:2008.04422](#) | [DOI:10.1007/JHEP09\(2020\)149](#) | 2020-08-10

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[JHEP 01 \(2021\) 163](#) | [arXiv:2007.05658](#) | [DOI:10.1007/JHEP01\(2021\)163](#) | 2020-07-10

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Sirunyan, Albert M et al.

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Sirunyan, Albert M et al.

[JHEP 04 \(2019\) 031](#) | [arXiv:1810.05905](#) | [DOI:10.1007/JHEP04\(2019\)031](#) | 2018-10-13

678. Centrality and pseudorapidity dependence of the transverse energy density in pPb collisions at $\sqrt{s_{NN}} = 5.02$ TeV

Sirunyan, Albert M et al.

[Phys.Rev.C 100 \(2019\) 024902](#) | [arXiv:1810.05745](#) | [DOI:10.1103/PhysRevC.100.024902](#) | 2018-10-12

679. Evidence for light-by-light scattering and searches for axion-like particles in ultraperipheral PbPb collisions at $\sqrt{s_{NN}} = 5.02$ TeV

Sirunyan, Albert M et al.

[Phys.Lett.B 797 \(2019\) 134826](#) | [arXiv:1810.04602](#) | [DOI:10.1016/j.physletb.2019.134826](#) | 2018-10-10

680. Search for top quark partners with charge $5/3$ in the same-sign dilepton and single-lepton final states in proton-proton collisions at $\sqrt{s} = 13$ TeV

Sirunyan, Albert M et al.

[JHEP 03 \(2019\) 082](#) | [arXiv:1810.03188](#) | [DOI:10.1007/JHEP03\(2019\)082](#) | 2018-10-07

681. Measurement of B^0_s meson production in pp and PbPb collisions at $\sqrt{s_{NN}} = 5.02$ TeV

Sirunyan, Albert M et al.

[Phys.Lett.B 796 \(2019\) 168](#) | [arXiv:1810.03022](#) | [DOI:10.1016/j.physletb.2019.07.014](#) | 2018-10-06

682. Observation of prompt J/ψ meson elliptic flow in high-multiplicity pPb collisions at $\sqrt{s_{NN}} = 8.16$ TeV

Sirunyan, Albert M et al.

[Phys.Lett.B 791 \(2019\) 172](#) | [arXiv:1810.01473](#) | [DOI:10.1016/j.physletb.2019.02.018](#) | 2018-10-02

683. Search for new physics in final states with a single photon and missing transverse momentum in proton-proton collisions at $\sqrt{s} = 13$ TeV

Sirunyan, Albert M et al.

[JHEP 02 \(2019\) 074](#) | [arXiv:1810.00196](#) | [DOI:10.1007/JHEP02\(2019\)074](#) | 2018-09-29

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Sirunyan, Albert M. et al.

[Eur.Phys.J.C 79 \(2019\) 277](#) | [arXiv:1809.11080](#) | [DOI:10.1140/epjc/s10052-019-6774-8](#) | 2018-09-28

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Sirunyan, Albert M et al.

[Eur.Phys.J.C 79 \(2019\) 421](#) | [arXiv:1809.10733](#) | [DOI:10.1140/epjc/s10052-019-6909-y](#) | 2018-09-27

686. Search for single production of vector-like quarks decaying to a top quark and a W boson in proton-proton collisions at $\sqrt{s} = 13$ TeV

Sirunyan, Albert M et al.

[Eur.Phys.J.C 79 \(2019\) 90](#) | [arXiv:1809.08597](#) | [DOI:10.1140/epjc/s10052-019-6556-3](#) | 2018-09-23

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Sirunyan, Albert M et al.

[Phys.Rev.Lett. 122 \(2019\) 152001](#) | [arXiv:1809.08602](#) | [DOI:10.1103/PhysRevLett.122.152001](#) | 2018-09-23

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Sirunyan, Albert M et al.

[Phys.Lett.B 793 \(2019\) 520](#) | [arXiv:1809.05937](#) | [DOI:10.1016/j.physletb.2019.04.025](#) | 2018-09-16

689. Search for leptoquarks coupled to third-generation quarks in proton-proton collisions at $\sqrt{s}=13$ TeV

Sirunyan, Albert M et al.

[Phys.Rev.Lett. 121 \(2018\) 241802](#) | [arXiv:1809.05558](#) | [DOI:10.1103/PhysRevLett.121.241802](#) | 2018-09-14

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Sirunyan, Albert M et al.

[Eur.Phys.J.C 78 \(2018\) 939](#) | [arXiv:1809.03578](#) | [DOI:10.1140/epjc/s10052-018-6390-z](#) | 2018-09-10

691. Search for the associated production of the Higgs boson and a vector boson in proton-proton collisions at $\sqrt{s}=13$ TeV via Higgs boson decays to τ leptons

Sirunyan, Albert M et al.

[JHEP 06 \(2019\) 093](#) | [arXiv:1809.03590](#) | [DOI:10.1007/JHEP06\(2019\)093](#) | 2018-09-10

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Sirunyan, A.M. et al.

[JINST 13 \(2018\) P10005](#) | [arXiv:1809.02816](#) | [DOI:10.1088/1748-0221/13/10/P10005](#) | 2018-09-08

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[Phys.Rev.D 98 \(2018\) 092001](#) | [arXiv:1809.00327](#) | [DOI:10.1103/PhysRevD.98.092001](#) | 2018-09-02

694. Charged-particle nuclear modification factors in XeXe collisions at $\sqrt{s_{\text{NN}}} = 5.44$ TeV
Sirunyan, Albert M et al.
[JHEP 10 \(2018\) 138](#) | [arXiv:1809.00201](#) | [DOI:10.1007/JHEP10\(2018\)138](#) | 2018-09-01
695. Observation of Higgs boson decay to bottom quarks
Sirunyan, A. M. et al.
[Phys.Rev.Lett. 121 \(2018\) 121801](#) | [arXiv:1808.08242](#) | [DOI:10.1103/PhysRevLett.121.121801](#) | 2018-08-24
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Sirunyan, A. M. et al.
[Phys.Rev.D 98 \(2018\) 092014](#) | [arXiv:1808.07340](#) | [DOI:10.1103/PhysRevD.98.092014](#) | 2018-08-22
697. Search for a charged Higgs boson decaying to charm and bottom quarks in proton-proton collisions at $\sqrt{s} = 8$ TeV
Sirunyan, Albert M et al.
[JHEP 11 \(2018\) 115](#) | [arXiv:1808.06575](#) | [DOI:10.1007/JHEP11\(2018\)115](#) | 2018-08-20
698. Search for pair production of second-generation leptoquarks at $\sqrt{s} = 13$ TeV
Sirunyan, Albert M et al.
[Phys.Rev.D 99 \(2019\) 032014](#) | [arXiv:1808.05082](#) | [DOI:10.1103/PhysRevD.99.032014](#) | 2018-08-15
699. Search for an L_μ - L_τ gauge boson using $Z \rightarrow 4\mu$ events in proton-proton collisions at $\sqrt{s} = 13$ TeV
Sirunyan, Albert M et al.
[Phys.Lett.B 792 \(2019\) 345](#) | [arXiv:1808.03684](#) | [DOI:10.1016/j.physletb.2019.01.072](#) | 2018-08-10
700. Search for pair-produced resonances decaying to quark pairs in proton-proton collisions at $\sqrt{s} = 13$ TeV
Sirunyan, Albert M et al.
[Phys.Rev.D 98 \(2018\) 112014](#) | [arXiv:1808.03124](#) | [DOI:10.1103/PhysRevD.98.112014](#) | 2018-08-09
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Sirunyan, Albert M et al.
[Phys.Rev.D 98 \(2018\) 092011](#) | [arXiv:1808.03078](#) | [DOI:10.1103/PhysRevD.98.092011](#) | 2018-08-09
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[Phys.Rev.Lett. 121 \(2018\) 221802](#) | [arXiv:1808.02913](#) | [DOI:10.1103/PhysRevLett.121.221802](#) | 2018-08-08
703. Search for resonances in the mass spectrum of muon pairs produced in association with b quark jets in proton-proton collisions at $\sqrt{s} = 8$ and 13 TeV
Sirunyan, A.M. et al.
[JHEP 11 \(2018\) 161](#) | [arXiv:1808.01890](#) | [DOI:10.1007/JHEP11\(2018\)161](#) | 2018-08-06
704. Search for production of Higgs boson pairs in the four b quark final state using large-area jets in proton-proton collisions at $\sqrt{s} = 13$ TeV
Sirunyan, Albert M et al.
[JHEP 01 \(2019\) 040](#) | [arXiv:1808.01473](#) | [DOI:10.1007/JHEP01\(2019\)040](#) | 2018-08-04
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Sirunyan, Albert M et al.
[Phys.Rev.Lett. 122 \(2019\) 081804](#) | [arXiv:1808.01257](#) | [DOI:10.1103/PhysRevLett.122.081804](#) | 2018-08-03
706. Search for heavy resonances decaying into two Higgs bosons or into a Higgs boson and a W or Z boson in proton-proton collisions at 13 TeV

Sirunyan, Albert M et al.

[JHEP 01 \(2019\) 051](#) | [arXiv:1808.01365](#) | [DOI:10.1007/JHEP01\(2019\)051](#) | 2018-08-03

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[Phys.Lett.B 792 \(2019\) 107](#) | [arXiv:1807.11421](#) | [DOI:10.1016/j.physletb.2019.01.069](#) | 2018-07-30

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Sirunyan, Albert M. et al.

[JHEP 11 \(2018\) 079](#) | [arXiv:1807.07799](#) | [DOI:10.1007/JHEP11\(2018\)079](#) | 2018-07-20

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[Phys.Rev.Lett. 122 \(2019\) 021801](#) | [arXiv:1807.06325](#) | [DOI:10.1103/PhysRevLett.122.021801](#) | 2018-07-17

710. Search for dark matter particles produced in association with a top quark pair at $\sqrt{s} = 13$ TeV

Sirunyan, Albert M. et al.

[Phys.Rev.Lett. 122 \(2019\) 011803](#) | [arXiv:1807.06522](#) | [DOI:10.1103/PhysRevLett.122.011803](#) | 2018-07-17

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Sirunyan, Albert M. et al.

[JHEP 11 \(2018\) 113](#) | [arXiv:1807.05974](#) | [DOI:10.1007/JHEP11\(2018\)113](#) | 2018-07-16

712. Measurement of inclusive and differential Higgs boson production cross sections in the diphoton decay channel in proton-proton collisions at $\sqrt{s} = 13$ TeV

Sirunyan, Albert M. et al.

[JHEP 01 \(2019\) 183](#) | [arXiv:1807.03825](#) | [DOI:10.1007/JHEP01\(2019\)183](#) | 2018-07-10

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Sirunyan, Albert M et al.

[JINST 13 \(2018\) P10034](#) | [arXiv:1807.03289](#) | [DOI:10.1088/1748-0221/13/10/P10034](#) | 2018-07-09

714. Search for heavy resonances decaying into a vector boson and a Higgs boson in final states with charged leptons, neutrinos and b quarks at $\sqrt{s} = 13$ TeV

Sirunyan, Albert M. et al.

[JHEP 11 \(2018\) 172](#) | [arXiv:1807.02826](#) | [DOI:10.1007/JHEP11\(2018\)172](#) | 2018-07-08

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Sirunyan, Albert M. et al.

[Eur.Phys.J.C 79 \(2019\) 123](#) | [arXiv:1807.02810](#) | [DOI:10.1140/epjc/s10052-019-6620-z](#) | 2018-07-08

716. Search for supersymmetry in events with a τ lepton pair and missing transverse momentum in proton-proton collisions at $\sqrt{s} = 13$ TeV

Sirunyan, Albert M. et al.

[JHEP 11 \(2018\) 151](#) | [arXiv:1807.02048](#) | [DOI:10.1007/JHEP11\(2018\)151](#) | 2018-07-05

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Sirunyan, Albert M et al.

[Eur.Phys.J.C 79 \(2019\) 20](#) | [arXiv:1807.00782](#) | [DOI:10.1140/epjc/s10052-018-6482-9](#) | 2018-07-02

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Sirunyan, Albert M. et al.

[Eur.Phys.J.C 78 \(2018\) 697](#) | [arXiv:1806.11245](#) | [DOI:10.1140/epjc/s10052-018-6144-y](#) | 2018-06-29

719. Measurement of differential cross sections for Z boson pair production in association with jets at $\sqrt{s} = 8$ and 13 TeV

Sirunyan, Albert M. et al.

[Phys.Lett.B 789 \(2019\) 19](#) | [arXiv:1806.11073](#) | [DOI:10.1016/j.physletb.2018.11.007](#) | 2018-06-28

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Sirunyan, Albert M et al.

[JHEP 01 \(2019\) 122](#) | [arXiv:1806.10905](#) | [DOI:10.1007/JHEP01\(2019\)122](#) | 2018-06-28

721. Search for the decay of a Higgs boson in the $\ell\ell\gamma$ channel in proton-proton collisions at $\sqrt{s} = 13$ TeV

Sirunyan, Albert M. et al.

[JHEP 11 \(2018\) 152](#) | [arXiv:1806.05996](#) | [DOI:10.1007/JHEP11\(2018\)152](#) | 2018-06-15

722. Search for supersymmetric partners of electrons and muons in proton-proton collisions at $\sqrt{s}=13$ TeV

Sirunyan, Albert M. et al.

[Phys.Lett.B 790 \(2019\) 140](#) | [arXiv:1806.05264](#) | [DOI:10.1016/j.physletb.2019.01.005](#) | 2018-06-13

723. Measurements of properties of the Higgs boson decaying to a W boson pair in pp collisions at $\sqrt{s}=13$ TeV

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[Phys.Lett.B 791 \(2019\) 96](#) | [arXiv:1806.05246](#) | [DOI:10.1016/j.physletb.2018.12.073](#) | 2018-06-13

724. Search for dark matter produced in association with a Higgs boson decaying to $\gamma\gamma$ or $\tau^+\tau^-$ at $\sqrt{s} = 13$ TeV

Sirunyan, Albert M. et al.

[JHEP 09 \(2018\) 046](#) | [arXiv:1806.04771](#) | [DOI:10.1007/JHEP09\(2018\)046](#) | 2018-06-12

725. Observation of the $Z\rightarrow\psi\ell^+\ell^-$ decay in pp collisions at $\sqrt{s}=13$ TeV

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[Phys.Rev.Lett. 121 \(2018\) 141801](#) | [arXiv:1806.04213](#) | [DOI:10.1103/PhysRevLett.121.141801](#) | 2018-06-11

726. Search for resonant pair production of Higgs bosons decaying to bottom quark-antiquark pairs in proton-proton collisions at 13 TeV

Sirunyan, Albert M. et al.

[JHEP 08 \(2018\) 152](#) | [arXiv:1806.03548](#) | [DOI:10.1007/JHEP08\(2018\)152](#) | 2018-06-09

727. Search for a singly produced third-generation scalar leptoquark decaying to a τ lepton and a bottom quark in proton-proton collisions at $\sqrt{s} = 13$ TeV

Sirunyan, Albert M. et al.

[JHEP 07 \(2018\) 115](#) | [arXiv:1806.03472](#) | [DOI:10.1007/JHEP07\(2018\)115](#) | 2018-06-09

728. Search for pair-produced resonances each decaying into at least four quarks in proton-proton collisions at $\sqrt{s}=13$ TeV

Sirunyan, Albert M. et al.

[Phys.Rev.Lett. 121 \(2018\) 141802](#) | [arXiv:1806.01058](#) | [DOI:10.1103/PhysRevLett.121.141802](#) | 2018-06-04

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Sirunyan, Albert M. et al.

[Eur.Phys.J.C 78 \(2018\) 701](#) | [arXiv:1806.00863](#) | [DOI:10.1140/epjc/s10052-018-6148-7](#) | 2018-06-03

730. Search for narrow and broad dijet resonances in proton-proton collisions at $\sqrt{s}=13$ TeV and constraints on dark matter mediators and other new particles

Sirunyan, Albert M et al.

[JHEP 08 \(2018\) 130](#) | [arXiv:1806.00843](#) | [DOI:10.1007/JHEP08\(2018\)130](#) | 2018-06-03

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Sirunyan, Albert M et al.

[Phys.Rev.D 98 \(2018\) 112011](#) | [arXiv:1806.00636](#) | [DOI:10.1103/PhysRevD.98.112011](#) | 2018-06-02

732. Search for Higgs boson pair production in the $\gamma\gamma b\bar{b}$ final state in pp collisions at $\sqrt{s}=13$ TeV

Sirunyan, Albert M et al.

[Phys.Lett.B 788 \(2019\) 7](#) | [arXiv:1806.00408](#) | [DOI:10.1016/j.physletb.2018.10.056](#) | 2018-06-01

733. Search for beyond the standard model Higgs bosons decaying into a $b\bar{b}$ pair in pp collisions at $\sqrt{s} = 13$ TeV

Sirunyan, Albert M et al.

[JHEP 08 \(2018\) 113](#) | [arXiv:1805.12191](#) | [DOI:10.1007/JHEP08\(2018\)113](#) | 2018-05-30

734. Observation of the $\chi_{b1}(3P)$ and $\chi_{b2}(3P)$ and measurement of their masses

Sirunyan, A. M. et al.

[Phys.Rev.Lett. 121 \(2018\) 092002](#) | [arXiv:1805.11192](#) | [DOI:10.1103/PhysRevLett.121.092002](#) | 2018-05-28

735. Constraints on models of scalar and vector leptoquarks decaying to a quark and a neutrino at $\sqrt{s}=13$ TeV

Sirunyan, Albert M et al.

[Phys.Rev.D 98 \(2018\) 032005](#) | [arXiv:1805.10228](#) | [DOI:10.1103/PhysRevD.98.032005](#) | 2018-05-25

736. Search for an exotic decay of the Higgs boson to a pair of light pseudoscalars in the final state with two b quarks and two τ leptons in proton-proton collisions at $\sqrt{s}=13$ TeV

Sirunyan, Albert M et al.

[Phys.Lett.B 785 \(2018\) 462](#) | [arXiv:1805.10191](#) | [DOI:10.1016/j.physletb.2018.08.057](#) | 2018-05-25

737. Measurement of nuclear modification factors of $\Upsilon(1S)$, $\Upsilon(2S)$, and $\Upsilon(3S)$ mesons in PbPb collisions at $\sqrt{s_{NN}} = 5.02$ TeV

Sirunyan, Albert M et al.

[Phys.Lett.B 790 \(2019\) 270](#) | [arXiv:1805.09215](#) | [DOI:10.1016/j.physletb.2019.01.006](#) | 2018-05-23

738. Measurement of the production cross section for single top quarks in association with W bosons in proton-proton collisions at $\sqrt{s}=13$ TeV

Sirunyan, Albert M et al.

[JHEP 10 \(2018\) 117](#) | [arXiv:1805.07399](#) | [DOI:10.1007/JHEP10\(2018\)117](#) | 2018-05-18

739. Search for black holes and sphalerons in high-multiplicity final states in proton-proton collisions at $\sqrt{s}=13$ TeV

Sirunyan, Albert M et al.

[JHEP 11 \(2018\) 042](#) | [arXiv:1805.06013](#) | [DOI:10.1007/JHEP11\(2018\)042](#) | 2018-05-15

740. Search for top squarks decaying via four-body or chargino-mediated modes in single-lepton final states in proton-proton collisions at $\sqrt{s} = 13$ TeV

Sirunyan, A.M. et al.

[JHEP 09 \(2018\) 065](#) | [arXiv:1805.05784](#) | [DOI:10.1007/JHEP09\(2018\)065](#) | 2018-05-15

741. Measurement of the groomed jet mass in PbPb and pp collisions at $\sqrt{s_{NN}}=5.02$ TeV

Sirunyan, Albert M et al.

[JHEP 10 \(2018\) 161](#) | [arXiv:1805.05145](#) | [DOI:10.1007/JHEP10\(2018\)161](#) | 2018-05-14

742. Search for an exotic decay of the Higgs boson to a pair of light pseudoscalars in the final state of two muons and two τ leptons in proton-proton collisions at $\sqrt{s}=13$ TeV

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[JHEP 11 \(2018\) 018](#) | [arXiv:1805.04865](#) | [DOI:10.1007/JHEP11\(2018\)018](#) | 2018-05-13

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[Phys.Rev.Lett. 121 \(2018\) 062002](#) | [arXiv:1805.04736](#) | [DOI:10.1103/PhysRevLett.121.062002](#) | 2018-05-12

744. Search for vector-like T and B quark pairs in final states with leptons at $\sqrt{s} = 13$ TeV

Sirunyan, Albert M et al.

[JHEP 08 \(2018\) 177](#) | [arXiv:1805.04758](#) | [DOI:10.1007/JHEP08\(2018\)177](#) | 2018-05-12

745. Measurement of prompt $\psi(2S)$ production cross sections in proton-lead and proton-proton collisions at $\sqrt{s_{NN}}= 5.02$ TeV

Sirunyan, Albert M et al.

[Phys.Lett.B 790 \(2019\) 509](#) | [arXiv:1805.02248](#) | [DOI:10.1016/j.physletb.2019.01.058](#) | 2018-05-06

746. Measurement of the top quark mass with lepton+jets final states using p p collisions at $\sqrt{s}=13$ TeV

Sirunyan, Albert M et al.

[Eur.Phys.J.C 78 \(2018\) 891](#) | [arXiv:1805.01428](#) | [DOI:10.1140/epjc/s10052-018-6332-9](#) | 2018-05-03

747. Elliptic flow of charm and strange hadrons in high-multiplicity pPb collisions at $\sqrt{s_{NN}} = 8.16$ TeV

Sirunyan, A. M. et al.

[Phys.Rev.Lett. 121 \(2018\) 082301](#) | [arXiv:1804.09767](#) | [DOI:10.1103/PhysRevLett.121.082301](#) | 2018-04-25

748. Search for disappearing tracks as a signature of new long-lived particles in proton-proton collisions at $\sqrt{s} = 13$ TeV

Sirunyan, Albert M et al.

[JHEP 08 \(2018\) 016](#) | [arXiv:1804.07321](#) | [DOI:10.1007/JHEP08\(2018\)016](#) | 2018-04-19

749. Measurement of differential cross sections for Z boson production in association with jets in proton-proton collisions at $\sqrt{s} = 13$ TeV

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[Eur.Phys.J.C 78 \(2018\) 965](#) | [arXiv:1804.05252](#) | [DOI:10.1140/epjc/s10052-018-6373-0](#) | 2018-04-14

750. Performance of the CMS muon detector and muon reconstruction with proton-proton collisions at $\sqrt{s}= 13$ TeV

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[JINST 13 \(2018\) P06015](#) | [arXiv:1804.04528](#) | [DOI:10.1088/1748-0221/13/06/P06015](#) | 2018-04-12

751. Search for ttH production in the $H \rightarrow b\bar{b}$ decay channel with leptonic tt decays in proton-proton collisions at $\sqrt{s}=13$ TeV

Sirunyan, Albert M et al.

[JHEP 03 \(2019\) 026](#) | [arXiv:1804.03682](#) | [DOI:10.1007/JHEP03\(2019\)026](#) | 2018-04-10

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Sirunyan, A.M. et al.

[JHEP 11 \(2018\) 185](#) | [arXiv:1804.02716](#) | [DOI:10.1007/JHEP11\(2018\)185](#) | 2018-04-08

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[Phys.Rev.Lett. 120 \(2018\) 231801](#) | [arXiv:1804.02610](#) | [DOI:10.1103/PhysRevLett.120.231801](#) | 2018-04-07

754. Search for a new scalar resonance decaying to a pair of Z bosons in proton-proton collisions at $\sqrt{s}=13$ TeV

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[JHEP 06 \(2018\) 127](#) | [arXiv:1804.01939](#) | [DOI:10.1007/JHEP06\(2018\)127](#) | 2018-04-05

755. Search for a heavy right-handed W boson and a heavy neutrino in events with two same-flavor leptons and two jets at $\sqrt{s}=13$ TeV

Sirunyan, Albert M et al.

[JHEP 05 \(2018\) 148](#) | [arXiv:1803.11116](#) | [DOI:10.1007/JHEP05\(2018\)148](#) | 2018-03-29

756. Search for high-mass resonances in final states with a lepton and missing transverse momentum at $\sqrt{s}=13$ TeV

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[JHEP 06 \(2018\) 128](#) | [arXiv:1803.11133](#) | [DOI:10.1007/JHEP06\(2018\)128](#) | 2018-03-29

757. Search for a heavy resonance decaying into a Z boson and a Z or W boson in $2\ell 2q$ final states at $\sqrt{s}=13$ TeV

Sirunyan, Albert M et al.

[JHEP 09 \(2018\) 101](#) | [arXiv:1803.10093](#) | [DOI:10.1007/JHEP09\(2018\)101](#) | 2018-03-27

758. Measurement of differential cross sections for the production of top quark pairs and of additional jets in lepton+jets events from pp collisions at $\sqrt{s}=13$ TeV

Sirunyan, Albert M et al.

[Phys.Rev.D 97 \(2018\) 112003](#) | [arXiv:1803.08856](#) | [DOI:10.1103/PhysRevD.97.112003](#) | 2018-03-23

759. Search for new physics in dijet angular distributions using proton-proton collisions at $\sqrt{s}=13$ TeV and constraints on dark matter and other models

Sirunyan, Albert M et al.

[Eur.Phys.J.C 78 \(2018\) 789](#) | [arXiv:1803.08030](#) | [DOI:10.1140/epjc/s10052-018-6242-x](#) | 2018-03-21

760. Search for ttH production in the all-jet final state in proton-proton collisions at $\sqrt{s}=13$ TeV

Sirunyan, Albert M et al.

[JHEP 06 \(2018\) 101](#) | [arXiv:1803.06986](#) | [DOI:10.1007/JHEP06\(2018\)101](#) | 2018-03-19

761. Search for additional neutral MSSM Higgs bosons in the $\tau\tau$ final state in proton-proton collisions at $\sqrt{s}=13$ TeV

Sirunyan, Albert M et al.

[JHEP 09 \(2018\) 007](#) | [arXiv:1803.06553](#) | [DOI:10.1007/JHEP09\(2018\)007](#) | 2018-03-17

762. Search for high-mass resonances in dilepton final states in proton-proton collisions at $\sqrt{s}=13$ TeV

Sirunyan, Albert M et al.

[JHEP 06 \(2018\) 120](#) | [arXiv:1803.06292](#) | [DOI:10.1007/JHEP06\(2018\)120](#) | 2018-03-16

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Sirunyan, Albert M et al.

[JHEP 08 \(2018\) 066](#) | [arXiv:1803.05485](#) | [DOI:10.1007/JHEP08\(2018\)066](#) | 2018-03-14

764. Observation of proton-tagged, central (semi)exclusive production of high-mass lepton pairs in pp collisions at 13 TeV with the CMS-TOTEM precision proton spectrometer

Sirunyan, Albert M et al.

[JHEP 07 \(2018\) 153](#) | [arXiv:1803.04496](#) | [DOI:10.1007/JHEP07\(2018\)153](#) | 2018-03-12

765. Measurements of differential cross sections of top quark pair production as a function of kinematic event variables in proton-proton collisions at $\sqrt{s}=13$ TeV

Sirunyan, A.M. et al.

[JHEP 06 \(2018\) 002](#) | [arXiv:1803.03991](#) | [DOI:10.1007/JHEP06\(2018\)002](#) | 2018-03-11

766. Search for a heavy resonance decaying into a Z boson and a vector boson in the $\nu\nu q\bar{q}$ final state

Sirunyan, Albert M et al.

[JHEP 07 \(2018\) 075](#) | [arXiv:1803.03838](#) | [DOI:10.1007/JHEP07\(2018\)075](#) | 2018-03-10

767. Search for third-generation scalar leptoquarks decaying to a top quark and a τ lepton at $\sqrt{s}=13$ TeV

Sirunyan, Albert M et al.

[Eur.Phys.J.C 78 \(2018\) 707](#) | [arXiv:1803.02864](#) | [DOI:10.1140/epjc/s10052-018-6143-z](#) | 2018-03-07

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Sirunyan, Albert M et al.

[JHEP 05 \(2018\) 006](#) | [arXiv:1803.00042](#) | [DOI:10.1007/JHEP05\(2018\)006](#) | 2018-02-28

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[JHEP 05 \(2018\) 088](#) | [arXiv:1802.09407](#) | [DOI:10.1007/JHEP05\(2018\)088](#) | 2018-02-26

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[Phys.Rev.Lett. 120 \(2018\) 201801](#) | [arXiv:1802.06149](#) | [DOI:10.1103/PhysRevLett.120.201801](#) | 2018-02-16

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Sirunyan, Albert M et al.

[Phys.Rev.D 97 \(2018\) 072010](#) | [arXiv:1802.04867](#) | [DOI:10.1103/PhysRevD.97.072010](#) | 2018-02-13

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[Phys.Rev.Lett. 120 \(2018\) 221801](#) | [arXiv:1802.02965](#) | [DOI:10.1103/PhysRevLett.120.221801](#) | 2018-02-08

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Sirunyan, Albert M et al.

[JHEP 07 \(2018\) 161](#) | [arXiv:1802.02613](#) | [DOI:10.1007/JHEP07\(2018\)161](#) | 2018-02-07

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Sirunyan, Albert M et al.

[JHEP 05 \(2018\) 025](#) | [arXiv:1802.02110](#) | [DOI:10.1007/JHEP05\(2018\)025](#) | 2018-02-06

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Sirunyan, Albert M et al.

[JHEP 06 \(2018\) 031](#) | [arXiv:1802.01486](#) | [DOI:10.1007/JHEP06\(2018\)031](#) | 2018-02-05

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Sirunyan, Albert M et al.

[JHEP 04 \(2018\) 073](#) | [arXiv:1802.01122](#) | [DOI:10.1007/JHEP04\(2018\)073](#) | 2018-02-04

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Sirunyan, Albert M et al.

[JHEP 03 \(2018\) 181](#) | [arXiv:1802.00707](#) | [DOI:10.1007/JHEP03\(2018\)181](#) | 2018-02-02

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[Phys.Rev.Lett. 121 \(2018\) 242301](#) | [arXiv:1801.04895](#) | [DOI:10.1103/PhysRevLett.121.242301](#) | 2018-01-15

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Sirunyan, Albert M et al.

[Eur.Phys.J.C 78 \(2018\) 708](#) | [arXiv:1801.03535](#) | [DOI:10.1140/epjc/s10052-018-6146-9](#) | 2018-01-10

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[Phys.Lett.B 782 \(2018\) 440](#) | [arXiv:1801.01846](#) | [DOI:10.1016/j.physletb.2018.05.062](#) | 2018-01-05

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Sirunyan, A.M. et al.

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Sirunyan, Albert M et al.

[Eur.Phys.J.C 78 \(2018\) 589](#) | [arXiv:1712.09814](#) | [DOI:10.1140/epjc/s10052-018-6049-9](#) | 2017-12-28

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[Phys.Lett.B 783 \(2018\) 114](#) | [arXiv:1712.08920](#) | [DOI:10.1016/j.physletb.2018.06.028](#) | 2017-12-24

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Sirunyan, Albert M. et al.

[Eur.Phys.J.C 78 \(2018\) 509](#) | [arXiv:1712.08959](#) | [DOI:10.1140/epjc/s10052-018-5950-6](#) | 2017-12-24

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[Phys.Rev.Lett. 120 \(2018\) 241801](#) | [arXiv:1712.08501](#) | [DOI:10.1103/PhysRevLett.120.241801](#) | 2017-12-22

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[JHEP 06 \(2018\) 001](#) | [arXiv:1712.07173](#) | [DOI:10.1007/JHEP06\(2018\)001](#) | 2017-12-19

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[Phys.Rev.C 97 \(2018\) 064912](#) | [arXiv:1712.07198](#) | [DOI:10.1103/PhysRevC.97.064912](#) | 2017-12-19

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[JINST 13 \(2018\) P05011](#) | [arXiv:1712.07158](#) | [DOI:10.1088/1748-0221/13/05/P05011](#) | 2017-12-19

789. Search for the $X(5568)$ state decaying into $B^0_{s1}\pi^\pm$ in proton-proton collisions at $\sqrt{s} = 8$ TeV

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[Phys.Rev.Lett. 120 \(2018\) 202005](#) | [arXiv:1712.06144](#) | [DOI:10.1103/PhysRevLett.120.202005](#) | 2017-12-17

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[Eur.Phys.J.C 78 \(2018\) 566](#) | [arXiv:1712.05471](#) | [DOI:10.1140/epjc/s10052-018-6033-4](#) | 2017-12-14

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[JHEP 09 \(2018\) 148](#) | [arXiv:1712.03143](#) | [DOI:10.1007/JHEP09\(2018\)148](#) | 2017-12-08

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[Phys.Lett.B 779 \(2018\) 358](#) | [arXiv:1712.02825](#) | [DOI:10.1016/j.physletb.2018.02.025](#) | 2017-12-07

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Sirunyan, Albert M et al.

[JHEP 06 \(2018\) 102](#) | [arXiv:1712.02399](#) | [DOI:10.1007/JHEP06\(2018\)102](#) | 2017-12-06

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Sirunyan, Albert M et al.

[JHEP 02 \(2018\) 032](#) | [arXiv:1712.02280](#) | [DOI:10.1007/JHEP02\(2018\)032](#) | 2017-12-06

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[Phys.Lett.B 778 \(2018\) 349](#) | [arXiv:1711.10949](#) | [DOI:10.1016/j.physletb.2018.01.049](#) | 2017-11-29

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[Phys.Lett.B 780 \(2018\) 432](#) | [arXiv:1711.09120](#) | [DOI:10.1016/j.physletb.2018.03.019](#) | 2017-11-24

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[Phys.Lett.B 780 \(2018\) 118](#) | [arXiv:1711.08008](#) | [DOI:10.1016/j.physletb.2018.02.045](#) | 2017-11-21

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[Phys.Lett.B 789 \(2019\) 643](#) | [arXiv:1711.05594](#) | [DOI:10.1016/j.physletb.2018.11.063](#) | 2017-11-15

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[Phys.Lett.B 781 \(2018\) 390](#) | [arXiv:1711.04652](#) | [DOI:10.1016/j.physletb.2018.04.007](#) | 2017-11-13
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Sirunyan, Albert M et al.
[JHEP 03 \(2018\) 003](#) | [arXiv:1711.04370](#) | [DOI:10.1007/JHEP03\(2018\)003](#) | 2017-11-12
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Sirunyan, A.M. et al.
[JHEP 07 \(2018\) 032](#) | [arXiv:1711.04299](#) | [DOI:10.1007/JHEP07\(2018\)032](#) | 2017-11-12
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Sirunyan, A.M. et al.
[JHEP 03 \(2018\) 115](#) | [arXiv:1711.03143](#) | [DOI:10.1007/JHEP03\(2018\)115](#) | 2017-11-08
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Sirunyan, Albert M et al.
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Sirunyan, A.M. et al.
[Eur.Phys.J.C 78 \(2018\) 287](#) | [arXiv:1711.02143](#) | [DOI:10.1140/epjc/s10052-018-5752-x](#) | 2017-11-06
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Sirunyan, Albert M et al.
[Phys.Rev.D 97 \(2018\) 032009](#) | [arXiv:1711.00752](#) | [DOI:10.1103/PhysRevD.97.032009](#) | 2017-11-02
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Sirunyan, A.M. et al.
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Sirunyan, A.M. et al.
[Phys.Lett.B 780 \(2018\) 251](#) | [arXiv:1710.11002](#) | [DOI:10.1016/j.physletb.2018.02.033](#) | 2017-10-30
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Sirunyan, Albert M et al.
[Phys.Rev.D 97 \(2018\) 012007](#) | [arXiv:1710.11188](#) | [DOI:10.1103/PhysRevD.97.012007](#) | 2017-10-30
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Sirunyan, Albert M et al.
[JHEP 01 \(2018\) 045](#) | [arXiv:1710.09355](#) | [DOI:10.1007/JHEP01\(2018\)045](#) | 2017-10-25

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[Eur.Phys.J.C 78 \(2018\) 457](#) | [arXiv:1710.08949](#) | [DOI:10.1140/epjc/s10052-018-5929-3](#) | 2017-10-24

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[Phys.Rev.C 98 \(2018\) 044902](#) | [arXiv:1710.07864](#) | [DOI:10.1103/PhysRevC.98.044902](#) | 2017-10-21

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[Eur.Phys.J.C 78 \(2018\) 242](#) | [arXiv:1710.02586](#) | [DOI:10.1140/epjc/s10052-018-5691-6](#) | 2017-10-06

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[Phys.Lett.B 780 \(2018\) 501](#) | [arXiv:1709.07497](#) | [DOI:10.1016/j.physletb.2018.02.050](#) | 2017-09-21

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[Phys.Rev.Lett. 120 \(2018\) 071802](#) | [arXiv:1709.05543](#) | [DOI:10.1103/PhysRevLett.120.071802](#) | 2017-09-16

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[JHEP 04 \(2018\) 033](#) | [arXiv:1709.05327](#) | [DOI:10.1007/JHEP04\(2018\)033](#) | 2017-09-15

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[JHEP 03 \(2018\) 166](#) | [arXiv:1709.05406](#) | [DOI:10.1007/JHEP03\(2018\)166](#) | 2017-09-15

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[Phys.Lett.B 779 \(2018\) 166](#) | [arXiv:1709.00384](#) | [DOI:10.1016/j.physletb.2017.12.069](#) | 2017-09-01

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[Phys.Rev.Lett. 120 \(2018\) 142302](#) | [arXiv:1708.09429](#) | [DOI:10.1103/PhysRevLett.120.142302](#) | 2017-08-30

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[Phys.Rev.C 100 \(2019\) 064908](#) | [arXiv:1708.08901](#) | [DOI:10.1103/PhysRevC.100.064908](#) | 2017-08-29

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[Phys.Lett.B 776 \(2018\) 195](#) | [arXiv:1702.00630](#) | [DOI:10.1016/j.physletb.2017.11.041](#) | 2017-02-02

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Khachatryan, Vardan et al.

[Eur.Phys.J.C 76 \(2016\) 325](#) | [arXiv:1601.04768](#) | [DOI:10.1140/epjc/s10052-016-4156-z](#) | 2016-01-18

1007. Measurement of inclusive jet production and nuclear modifications in pPb collisions at $\sqrt{s_{NN}} = 5.02$ TeV

Khachatryan, Vardan et al.

[Eur.Phys.J.C 76 \(2016\) 372](#) | [arXiv:1601.02001](#) | [DOI:10.1140/epjc/s10052-016-4205-7](#) | 2016-01-08

1008. Correlations between jets and charged particles in PbPb and pp collisions at $\sqrt{s_{NN}}=2.76$ TeV

Khachatryan, Vardan et al.

[JHEP 02 \(2016\) 156](#) | [arXiv:1601.00079](#) | [DOI:10.1007/JHEP02\(2016\)156](#) | 2016-01-01

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Khachatryan, Vardan et al.

[JHEP 04 \(2016\) 005](#) | [arXiv:1512.08377](#) | [DOI:10.1007/JHEP04\(2016\)005](#) | 2015-12-28

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Khachatryan, Vardan et al.

[Phys.Lett.B 759 \(2016\) 9](#) | [arXiv:1512.08002](#) | [DOI:10.1016/j.physletb.2016.05.033](#) | 2015-12-25

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Khachatryan, Vardan et al.

[Phys.Lett.B 759 \(2016\) 36](#) | [arXiv:1512.06461](#) | [DOI:10.1016/j.physletb.2016.05.044](#) | 2015-12-20

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Khachatryan, Vardan et al.

[Eur.Phys.J.C 76 \(2016\) 265](#) | [arXiv:1512.06212](#) | [DOI:10.1140/epjc/s10052-016-4083-z](#) | 2015-12-19

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Khachatryan, Vardan et al.

[Phys.Rev.Lett. 116 \(2016\) 071801](#) | [arXiv:1512.01224](#) | [DOI:10.1103/PhysRevLett.116.071801](#) | 2015-12-03

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Khachatryan, Vardan et al.

[Eur.Phys.J.C 76 \(2016\) 155](#) | [arXiv:1512.00815](#) | [DOI:10.1140/epjc/s10052-016-3988-x](#) | 2015-12-02

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Khachatryan, Vardan et al.

[Phys.Rev.D 93 \(2016\) 052011](#) | [arXiv:1511.09375](#) | [DOI:10.1103/PhysRevD.93.052011](#) | 2015-11-30

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Khachatryan, Vardan et al.

[Phys.Lett.B 758 \(2016\) 321](#) | [arXiv:1511.06170](#) | [DOI:10.1016/j.physletb.2016.05.005](#) | 2015-11-19

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Khachatryan, Vardan et al.

[JHEP 04 \(2016\) 035](#) | [arXiv:1511.03951](#) | [DOI:10.1007/JHEP04\(2016\)035](#) | 2015-11-12

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Khachatryan, Vardan et al.

[Phys.Lett.B 758 \(2016\) 296](#) | [arXiv:1511.03610](#) | [DOI:10.1016/j.physletb.2016.05.003](#) | 2015-11-11

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Khachatryan, Vardan et al.

[JHEP 04 \(2016\) 073](#) | [arXiv:1511.02138](#) | [DOI:10.1007/JHEP04\(2016\)073](#) | 2015-11-06

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Khachatryan, Vardan et al.

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Khachatryan, Vardan et al.

[JINST 11 \(2016\) P01019](#) | [arXiv:1510.07488](#) | [DOI:10.1088/1748-0221/11/01/P01019](#) | 2015-10-26

1022. Search for a very light NMSSM Higgs boson produced in decays of the 125 GeV scalar boson and decaying into τ leptons in pp collisions at $\sqrt{s} = 8$ TeV

Khachatryan, Vardan et al.

[JHEP 01 \(2016\) 079](#) | [arXiv:1510.06534](#) | [DOI:10.1007/JHEP01\(2016\)079](#) | 2015-10-22

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Khachatryan, Vardan et al.

[Phys.Rev.Lett. 116 \(2016\) 052002](#) | [arXiv:1510.05302](#) | [DOI:10.1103/PhysRevLett.116.052002](#) | 2015-10-18

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Khachatryan, Vardan et al.
[JHEP 12 \(2015\) 178](#) | [arXiv:1510.04252](#) | [DOI:10.1007/JHEP12\(2015\)178](#) | 2015-10-14
1025. Transverse momentum spectra of inclusive b jets in pPb collisions at $\sqrt{s_{NN}} = 5.02$ TeV
Khachatryan, Vardan et al.
[Phys.Lett.B 754 \(2016\) 59](#) | [arXiv:1510.03373](#) | [DOI:10.1016/j.physletb.2016.01.010](#) | 2015-10-12
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Khachatryan, Vardan et al.
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Khachatryan, Vardan et al.
[Eur.Phys.J.C 76 \(2016\) 379](#) | [arXiv:1510.03072](#) | [DOI:10.1140/epjc/s10052-016-4105-x](#) | 2015-10-11
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Khachatryan, Vardan et al.
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Khachatryan, Vardan et al.
[Phys.Lett.B 755 \(2016\) 217](#) | [arXiv:1510.01181](#) | [DOI:10.1016/j.physletb.2016.01.056](#) | 2015-10-05
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Khachatryan, Vardan et al.
[JHEP 01 \(2016\) 166](#) | [arXiv:1509.08141](#) | [DOI:10.1007/JHEP01\(2016\)166](#) | 2015-09-27
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Khachatryan, Vardan et al.
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Khachatryan, Vardan et al.
[Eur.Phys.J.C 76 \(2016\) 128](#) | [arXiv:1509.06076](#) | [DOI:10.1140/epjc/s10052-016-3956-5](#) | 2015-09-20
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Khachatryan, Vardan et al.
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Khachatryan, Vardan et al.

[Phys.Rev.D 93 \(2016\) 012003](#) | [arXiv:1509.04177](#) | [DOI:10.1103/PhysRevD.93.012003](#) | 2015-09-14

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[Phys.Lett.B 759 \(2016\) 641](#) | [arXiv:1509.03893](#) | [DOI:10.1016/j.physletb.2016.06.027](#) | 2015-09-13

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[Phys.Rev.D 93 \(2016\) 032004](#) | [arXiv:1509.03744](#) | [DOI:10.1103/PhysRevD.93.032004](#) | 2015-09-12

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Khachatryan, Vardan et al.

[Phys.Rev.D 93 \(2016\) 032005](#) | [arXiv:1509.03750](#) | [DOI:10.1103/PhysRevD.93.032005](#) | 2015-09-12

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Khachatryan, Vardan et al.

[JHEP 11 \(2015\) 018](#) | [arXiv:1508.07774](#) | [DOI:10.1007/JHEP11\(2015\)018](#) | 2015-08-31

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[Eur.Phys.J.C 76 \(2016\) 13](#) | [arXiv:1508.07819](#) | [DOI:10.1140/epjc/s10052-015-3853-3](#) | 2015-08-31

1042. Search for supersymmetry in the vector-boson fusion topology in proton-proton collisions at $\sqrt{s} = 8$ TeV

Khachatryan, Vardan et al.

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Khachatryan, Vardan et al.

[Phys.Rev.Lett. 116 \(2016\) 032301](#) | [arXiv:1508.06678](#) | [DOI:10.1103/PhysRevLett.116.032301](#) | 2015-08-26

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Khachatryan, Vardan et al.

[Phys.Lett.B 755 \(2016\) 196](#) | [arXiv:1508.04308](#) | [DOI:10.1016/j.physletb.2016.02.002](#) | 2015-08-18

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Khachatryan, Vardan et al.

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Khachatryan, Vardan et al.
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Khachatryan, Vardan et al.
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Khachatryan, Vardan et al.
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Khachatryan, Vardan et al.
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Khachatryan, Vardan et al.
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Khachatryan, Vardan et al.
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Khachatryan, Vardan et al.
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Khachatryan, Vardan et al.
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Khachatryan, Vardan et al.
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Khachatryan, Vardan et al.
[Phys.Lett.B 753 \(2016\) 363](#) | [arXiv:1507.00359](#) | [DOI:10.1016/j.physletb.2015.12.017](#) | 2015-07-01
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Khachatryan, Vardan et al.
[JHEP 11 \(2015\) 071](#) | [arXiv:1506.08329](#) | [DOI:10.1007/JHEP11\(2015\)071](#) | 2015-06-27

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Khachatryan, Vardan et al.
[Phys.Rev.D 93 \(2016\) 012001](#) | [arXiv:1506.03062](#) | [DOI:10.1103/PhysRevD.93.012001](#) | 2015-06-09
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Khachatryan, Vardan et al.
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Khachatryan, Vardan et al.
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Khachatryan, V. et al.
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Khachatryan, Vardan et al.
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Khachatryan, Vardan et al.
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1068. Search for a pseudoscalar boson decaying into a Z boson and the 125 GeV Higgs boson in $\ell^+\ell^-b\bar{b}$ final states
Khachatryan, Vardan et al.
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1069. Angular coefficients of Z bosons produced in pp collisions at $\sqrt{s}=8$ TeV and decaying to $\mu^+\mu^-$ as a function of transverse momentum and rapidity
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1070. Measurement of the Z boson differential cross section in transverse momentum and rapidity in proton-proton collisions at 8 TeV
Khachatryan, Vardan et al.
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1071. Search for the production of dark matter in association with top-quark pairs in the single-lepton final state in proton-proton collisions at $\sqrt{s}=8$ TeV
Khachatryan, Vardan et al.
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Khachatryan, Vardan et al.

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1079. Search for resonant pair production of Higgs bosons decaying to two bottom quark-antiquark pairs in proton-proton collisions at 8 TeV

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1084. Measurement of the $Z\gamma$ Production Cross Section in pp Collisions at 8 TeV and Search for Anomalous Triple Gauge Boson Couplings

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[JINST 10 \(2015\) P08010](#) | [arXiv:1502.02702](#) | [DOI:10.1088/1748-0221/10/08/P08010](#) | 2015-02-09

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[JINST 10 \(2015\) P06005](#) | [arXiv:1502.02701](#) | [DOI:10.1088/1748-0221/10/06/P06005](#) | 2015-02-09

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[JINST 9 \(2014\) P10009](#) | [arXiv:1405.6569](#) | [DOI:10.1088/1748-0221/9/10/P10009](#) | 2014-05-26

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Chatrchyan, Serguei et al.

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